

About Sterling B2B Integrator

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Background

Introduction	This lesson provides you an overview on Sterling B2B Integrator.
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Lesson objectives	This lesson is designed to help you to: ■ List typical business integration problems ■ Explain the basics of Sterling B2B Integrator ■ List the benefits of Sterling B2B Integrator ■ Identify Sterling B2B Integrator user interfaces
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Instructor Note: Inform students that in this lesson we begin with the background on Sterling B2B Integrator and how to use this software in an environment.

Overview of Sterling B2B Integrator

What is Sterling B2B Integrator?

Sterling B2B Integrator is the next generation end-to-end enterprise integration server for dynamic real-time and near-real time business collaboration and commerce automation. It is a business process-centric transaction engine for modeling and managing processes. It handles complex routing, translation, flexible integration, and real-time interaction with multiple internal systems and external trading partners. Major highlights include:

- Powerful, advanced translation and routing capabilities for high performance EDI translation and complex XML transformations and filtering.
- Process engine that is optimized for high-speed synchronous and asynchronous complex multi-step message exchanges (XML-based, real-time EDI, B2B standards such as ebXML, and user-defined private processes)
- Component-based Java 2 Enterprise Edition (J2EE) architecture for distributed enterprise scaling, reliability, transaction management, and plug-in extensibility for emerging B2B standards and integration adapters.
- Layered business-process-centric architecture employs industry standard BPML meta-language; insulates (preserves) business process specifications from potential fluid low-level technologies.
- Integrated universal transformation capability of IBM WebSphere Transformation Extender with business process management features of Sterling B2B Integrator.
- Configured to transfer files in and out of WebSphere MQ File Transfer Edition Network.

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Overview of Sterling B2B Integrator (Continued)

Typical Business Process Problems

Business managers are under tremendous stress to reduce costs and increase efficiency in communication/information transfer between their companies and clients, as well as internally from department to department.

The lack of application integration, both internally and externally, results in many problems:

- Slow response time, leaving customers dissatisfied
- Reliance on excessive manual processing, increasing costs and human error
- Inability to adopt new standards, causing missed opportunities
- Restrictive architecture of systems, limiting expansion

While business managers struggle with these issues, IT Managers/Directors are also trying to streamline and improve business processes without incurring high expenses. In many cases, they are using legacy systems with high investments and do not want to abandon for a new system. However, they lack the time and staff to overhaul the existing system and need to have fast, dependable service from a trustworthy provider.

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Overview of Sterling B2B Integrator (Continued)

Benefits of Sterling B2B Integrator

Because Sterling B2B Integrator was built from the ground up to meet B2Bi needs, it provides:

- Simplified management
- Consistent integration deployment methodology
- Capability to model and deploy complex end-to-end processes

Every company wants application integration so they can meet their business goals. Some of these business goals are to:

- Reduce operational costs
 - Increase operational efficiencies
 - Improve customer satisfaction
 - Increase revenues
 - Retain or increase competitive position
 - Improve responsiveness
-

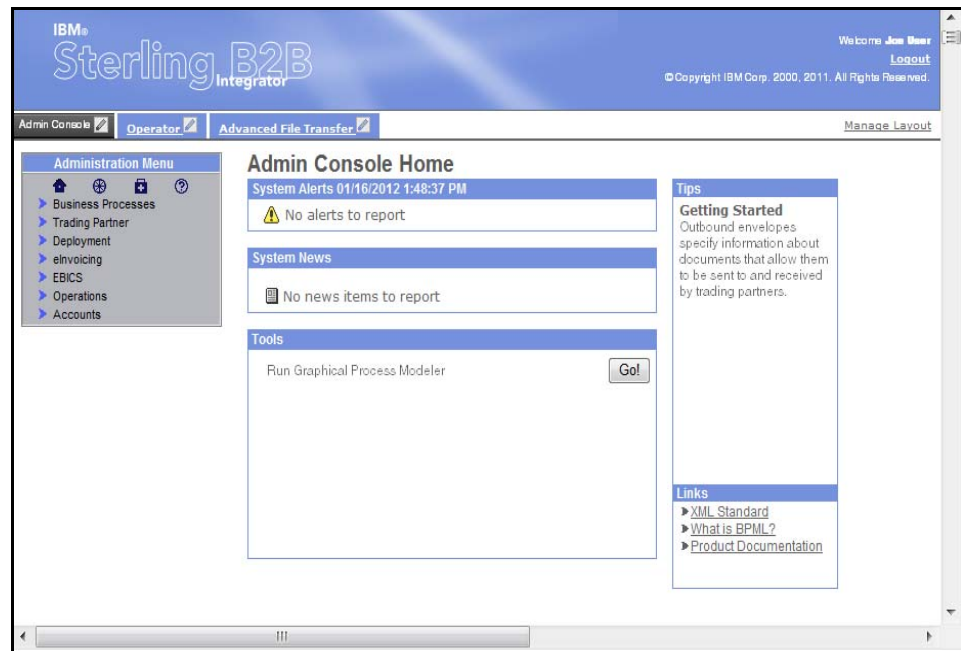
User Interfaces

Overview The three main user interfaces that are available within Sterling B2B Integrator are:

- The Administration Interface
- The Sterling B2B Integrator Map Editor
- The Graphical Process Modeler

The Administration Interface

The Administration Interface that is shown here is a web-based interface for System Administration.

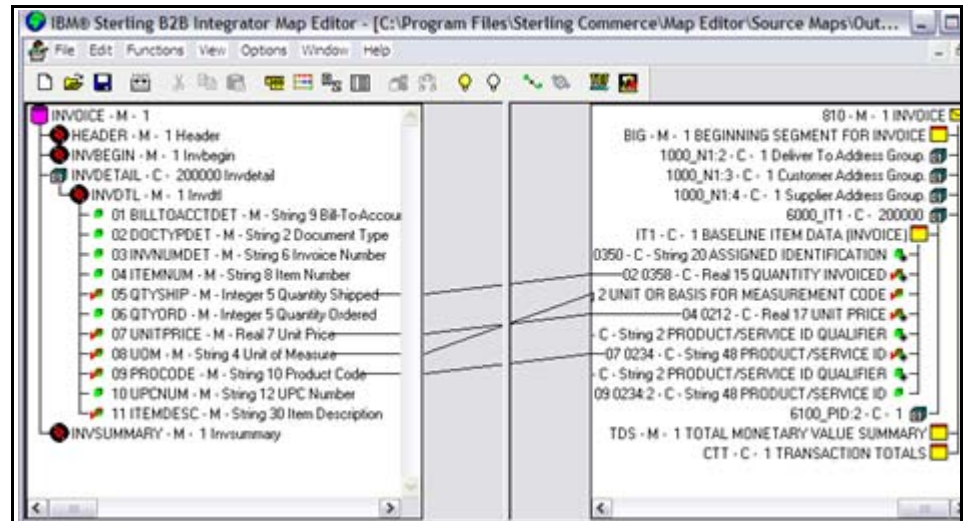


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User Interfaces

(Continued)**Overview
(Continued)****The Sterling B2B Integrator Map Editor**

The Sterling B2B Integrator Map Editor is shown in the following figure. It is used to represent the translation of one type of input data to a different type of output data.

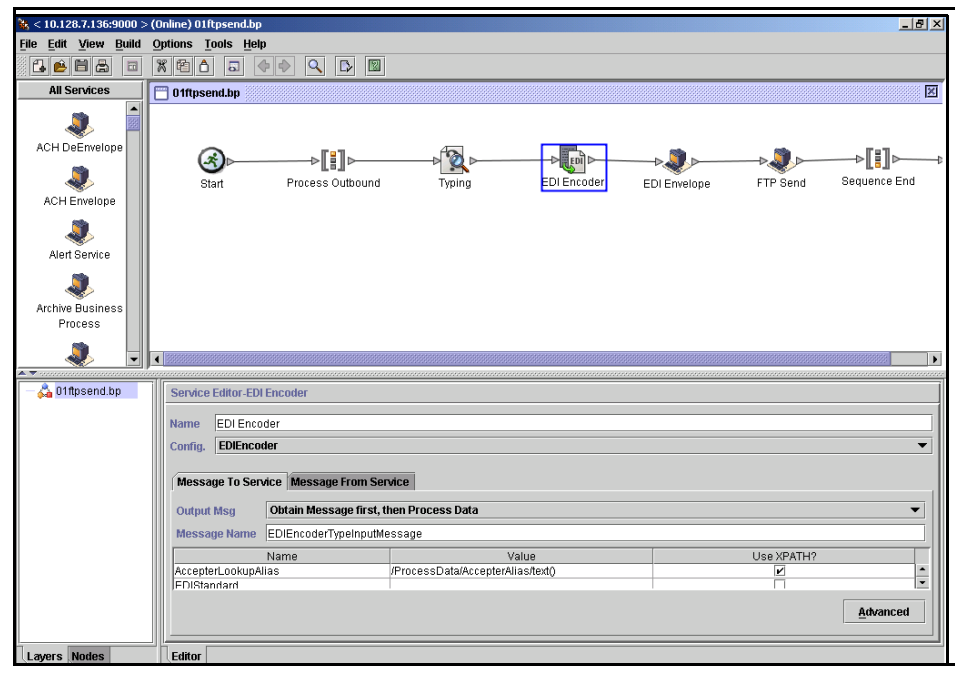


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User Interfaces

(Continued)**Overview
(Continued)****The Graphical Process Modeler**

The Graphical Process Modeler is shown in the following image. It is used to represent a business process as a graphical model.



Instructor Note: Use this time as a demo and navigation session - navigate through the system with them.

The Administrator Interface

Overview The Sterling B2B Integrator administration interface allows you to set up, configure, and maintain the system. There are six main areas in the Sterling B2B Integrator Administration Interface:

- Business Process
- Trading Partners
- Deployment
- eInvoicing
- Operations
- Accounts



Important!	When using browser buttons within Sterling B2B Integrator, do not use the back arrow or back button of Internet Explorer.
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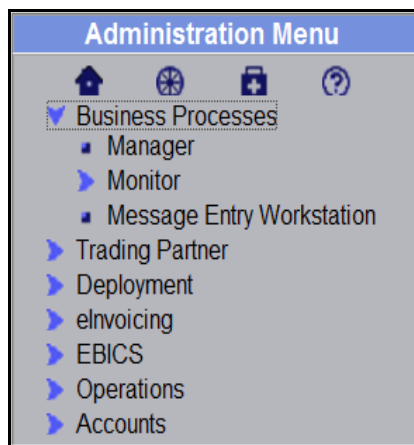
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The Administrator Interface

(Continued)

Business Processes

The Business Process menu of Sterling B2B Integrator allows you to manage and monitor business processes.



Business Process Manager

The Business Process Manager screen provides the options that you need to create and locate business process definitions. This screen is used to:

- Download Java WebStart
- Run the Graphical Process Modeler
- Create new process definitions
- Search for process definitions
- List process definitions

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The Administrator Interface

(Continued)

Business Processes (Continued)

Business Process Monitor - Current Processes

The Business Process Monitor -Current Processes screen displays a summary of the processes in Sterling B2B Integrator. Each color indicates the status of the business process:

- Green - No errors or warnings during processing
- Red - Errors or warnings that are encountered during processing

From this screen, you can:

- Review detailed processing information
- View business process definitions
- Obtain document information

Important!	If you are using Internet Explorer, this screen automatically refreshes every minute. If using Firefox or Chrome, you must manually refresh the screen. You can stop the automatic refresh activity by clearing the automatic refresh check box.
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The Administrator Interface

(Continued)

Business Processes (Continued)

Business Process Monitor Central Search screen

The Business Process Monitor Central Search screen provides you the ability to locate and view business processes based on specific search criteria:

- Business Process Name
- Status
- Start Date From
- Start Date To

Business Process Monitor Advanced Search screen

The Business Process Monitor Advanced Search screen provides you the ability to locate and view business processes based on specific search criteria:

- Location (Live tables, Archive tables, Restore tables)
- Business Process ID
- Business Process Name
- System Business Process Name
- State
- Status
- Start Date From
- Start Date To

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The Administrator Interface

(Continued)

Business Processes (Continued)

Business Process Monitor Current Activities Search screen

The Business Process Monitor Current Activities Search screen takes you to the Service Activity page. This page is used primarily to look at adapters and the status of large file transfers. This page provides you the ability to search for service activities with the following criteria:

- Service Type
- Service Name
- Service Activity Type
- Not Updated Since: Date/Time
- Started Before: Date/Time

Business Process Monitor Advanced Search - Documents Screen

The Business Process Monitor Advanced Search - Documents screen allows you to see the most recent documents that are processed without specifying any search options. Current documents give you the ability to monitor current document processing at a glance from the page in Sterling B2B Integrator. The Documents page provides you the ability to search for documents with the following criteria:

- Start Date From
- Sender ID
- Type
- Receiver ID
- Status
- Tracking ID

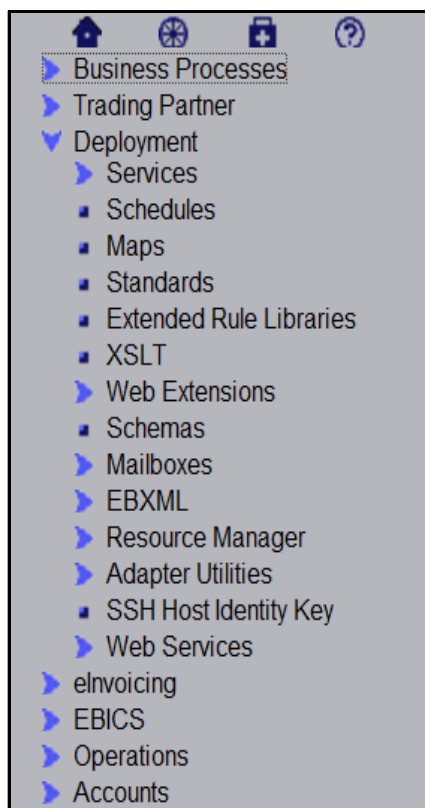
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The Administrator Interface

(Continued)

Deployment

The Deployment menu allows you to configure services and maps. Details of the Trading Partner, Operations, and Accounts menus is elaborated on in further classes.



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The Administrator Interface

(Continued)

Deployment (Continued)

Services

The Services menu provides options to install/setup and configure the services for Sterling B2B Integrator.

Schedules

The Schedules option allows you to schedule business processes to run at specified times. It also allows you to delete schedules without actually deleting a business process or a service.

Maps

The Maps option allows you to download/install the Map Editor, check in/out maps, and find maps.

Standards

Standards define data translation specifications for documents. The Standards Download page allows you to download and install the standards that are required for data translation the Map Editor.

Extended Rule Libraries

The Extended Rule Libraries page allows you to check in new or updated Extended Rule Libraries. You can also, search for an Extended Rule Library and list all Extended Rule Libraries or list the specific Extended Rule Libraries searched based on the first letter of the name.

XSLT

The XSLT option allows you to create and manipulate extensible style sheets.

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The Administrator Interface

(Continued)

Deployment (Continued)

Web Extensions

Web Extension allows you to download and install the **Web Template Translator** application; create, edit, check-in, and check-out web templates; and download and install web designer extensions to create XForms.

Schemas

The **Schema** screen allows you to load an XML schema or DTD into the system.

Mailboxes

The Mailboxes option allows you to organize, store, and manage trading partner documents and transactions.

EBXML

The EBXML option allows you to manage and locate Business Process Specification Schema (BPSS) and Collaboration Protocol Agreements (CPA).

Resource Manager

The Resource Manager allows you to create and locate resource tags, and import or export resources.

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The Administrator Interface

(Continued)

Deployment (Continued)

Adapter Utilities

The Adapter Utilities screen allows you to create DTDs and locate existing DTDs for Siebel Systems. This menu option is displayed in the Deployment menu if the respective service were already installed. Additionally, a Service Development Kit is available from this menu item.

SSH Host Identity Key

The SSH Host Identity Key allows you to create the necessary keys for SSH and SCP protocols.

Web Services

A Web Service Configuration allows you to locate and configure certain services and business processes to be exposed as web services.

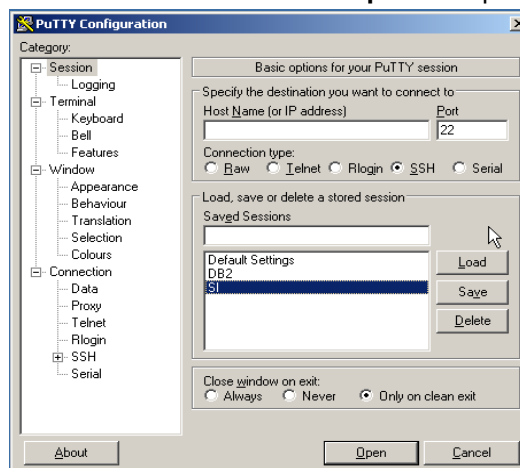
Instructor Note: Now, take a closer look at the components in Sterling B2B Integrator and explain how the Sterling B2B Integrator functions between the enterprise and the Trading Partners.

Exercise 1–1 Starting Sterling B2B Integrator

Instructions

You are required to start the Sterling B2B Integrator to open and view the Sterling B2B Integrator user interfaces. Follow the steps that are listed to start the Sterling B2B Integrator:

1. Open the windows image with the user name and password given to you.
2. Access Sterling B2B Integrator server through Putty.
3. Select **SI** in the **Saved Sessions** and click **Open**. The putty session opens.



4. Login as **root** with the password **root123**.
5. Go to the installation directory with the following command.

```
cd /opt/IBM/SterlingIntegrator/install/bin
```

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Instructor Note: Provide the students with the login details for windows image.

Exercise 1–1 Starting Sterling B2B Integrator (Continued)

Instructions

6. Execute the following command.
`./hardstop.sh`
7. On successful completion of the command, execute the following command.
`./run.sh`



Caution

B2Bi's database is set to autostart. In the VM environment sometimes it does not autostart. If after running `./run.sh` you get JDBC connection errors goto the DB2 Start section otherwise proceed to step 8

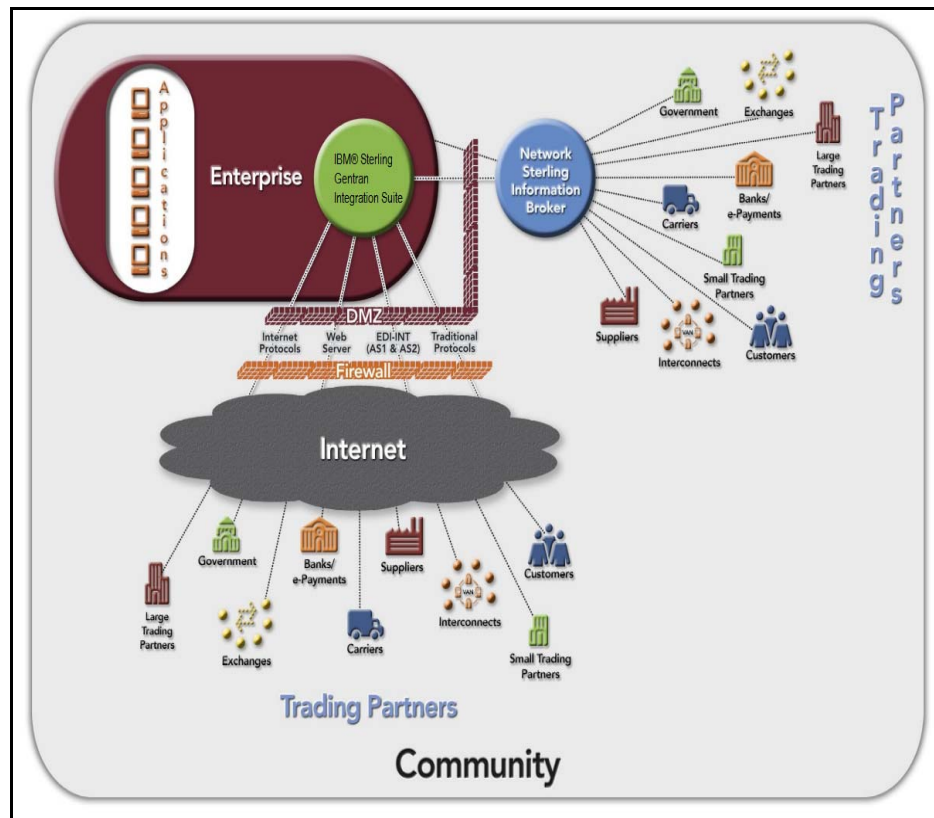
DB2 Start

1. From the same **PuTTY** window, enter `su db2inst1` and click **Enter** to switch the user from root to db2inst1.
 2. Enter `db2start`, and click **Enter**. Upon successful start, the message '<db2start> processing was successful' is displayed.
 3. Enter `exit` and click **Enter** to exit user db2inst1 and return to the root user. Enter `./run.sh` again to start B2Bi.
 8. Enter the passphrase as **password** when it prompts for a passphrase. A message indicating the successful start of Sterling B2B Integrator will be displayed.
 9. Connect to the Sterling B2B integrator Dashboard through the browser with the link `http://192.168.40.100:9000/dashboard/`.
 10. Sign in as **admin** with the password as **password**. The **Admin Console Home** page opens.
-

Components

Overview of Components

The following diagram illustrates the main components and communication paths between them.

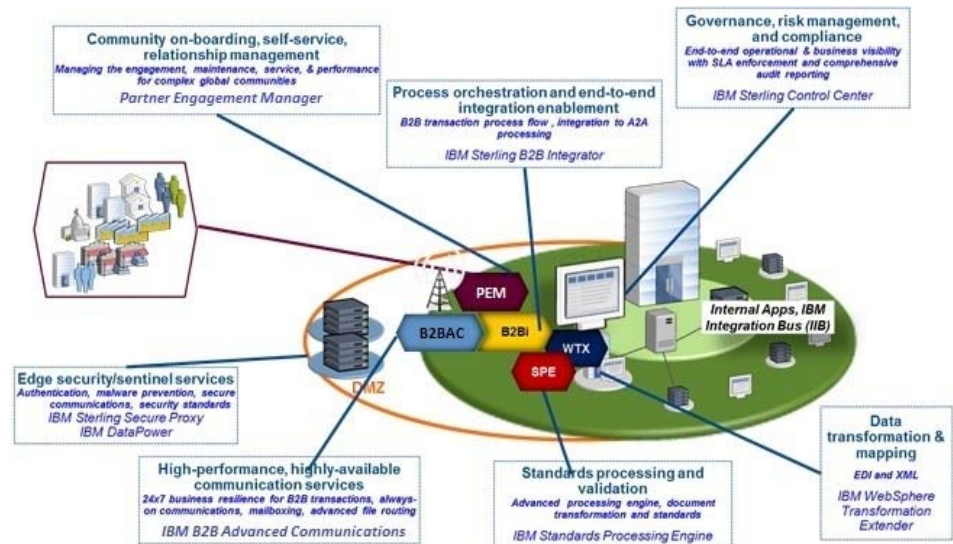


Instructor Note: Explain the different components and communication channels with the help of the diagram.

B2B Components

(Continued)

IBM B2B Strategy



IBM's B2B strategy includes the following Components:

Global Mailbox

IBM® Global High Availability Mailbox is a robust and reliable way to store message data from partners in any geography. As a component of IBM Sterling B2B Integrator, Global Mailbox enables customers who have licensed to optionally pay for an add-on component that provides high availability mailboxing to exchange files. This HA mailboxing capability allows clients to deploy a B2B platform that can sustain continuous operations plus offers differentiating disaster recovery capabilities.

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B2B Components

(Continued)

IBM B2B Strategy (Continued)

Global Mailbox provides data storage across geographically distributed locations. Data is routed to the nearest available server, and then replicated quickly across data centers so that the data is available even if a data center is not accessible. Applications that are enabled for Global Mailbox listen for mailbox events to trigger application level actions, such as to initiate further processing.

Partner Engagement Manager (PEM)

IBM Partner Engagement Manager (IBM PEM) is a multi-enterprise B2B engagement portal that enables collaboration, interaction, monitoring, and visibility of activity flows between sponsor organizations and their partners, customers, and suppliers.

ITX Advanced (Standards Processing Engine)

The Standards Processing Engine portfolio is the next evolution of the IBM® Universal Transformation strategy for transforming documents. It contains modular, comprehensive solutions that are based on industry standards. The solutions provide support for WebSphere® Transformation Extender, Sterling B2B Integrator, and XSLT translation maps, providing document processing that is modular, scalable, and extensible.

Control Center

IBM Control Center V6.0 is a centralized monitoring and management system that lets operations personnel continuously monitor business activities across the enterprise for IBM Sterling products and for many FTP servers and manage the configurations and licenses of Sterling Connect:Direct servers.

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B2B Components

(Continued)

IBM B2B Strategy (Continued)

Secure Proxy

Sterling Secure Proxy acts as an application proxy between Sterling Connect:Direct® nodes or between a client application and a Sterling B2B Integrator server. It provides a high level of data protection between external connections and your internal network. Define an inbound node definition for each trading partner connection from outside the company and an outbound node definition for every company server to which Sterling Secure Proxy will connect.

IBM Advanced Communications

B2B Advanced Communications provides secure, optimized, and dynamic end-to-end information flows supporting the viewing of a single exchange, configuring and managing of trading partners, and support for both AS2 and AS4 protocols

Lesson Review

Completed Objectives

This lesson was designed to enable you to:

- List typical business integration problems
 - Explain the basics of Sterling B2B Integrator
 - List the benefits of Sterling B2B Integrator
 - Identify Sterling B2B Integrator user interfaces
-

Basic BPML Concepts

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Background

Introduction	This lesson provides an overview of the components that make up the Business Process Modeling Language (BPML).
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Lesson objectives	This lesson is designed to help you to: ■ Define BPML ■ Discuss the activities that perform BPML functions ■ Differentiate between services and adapters as they relate to Sterling B2B Integrator ■ Explain how Sterling B2B Integrator uses BPML
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What is BPML?

Overview

BPML is short for Business Process Modeling Language. It is an **XML-based** language for describing a business process and telling it how to run. Not very much XML knowledge is required to comprehend BPML. You should know the basics of XML. For example, you must know about the elements and attributes, usage of start, end, and empty tags, and other basic XML concepts. Since it is based on XML it uses its components and basic structure but not its rules and additional capabilities.

An understanding of XPath (an XML Search language) is necessary to search and access document content. Advanced features such as DTD structure or various entity references are not needed. For more information about XPath, see Lesson 4, XPath Basics.

Services and Adapters

Services and adapters are software components that can be used to accomplish integration or business tasks. Business processes make calls to services and adapters to accomplish your business tasks. For a business process to interact with these components, you must include services as steps in your business process. A service is a component of Sterling B2B Integrator, such as a translator. Services that access disk or other applications (reach outside of Sterling B2B Integrator) are known as adapters.

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Primary Document/Process Data	Moving forward there are two main concepts that are crucial to your understanding of Sterling B2B Integrator and business processes. You do not have to have a complete understanding of them immediately but do start to pay attention to them as you are introduced to BPML and start building your own Business Processes.
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Table 2–1:

Process Data Acts as a virtual memory for the business process. Is always in XML Information can be assigned to it or retrieved from it to be used as variables in your business process.	Primary Document Document being worked on. Such as an EDI doc to be translated. Can be any type of data Is an object link in Process data. A service/adaptor will always work on Primary Document.
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(Continued)

What is BPML?

(Continued)

Messages

Message exchange

Message exchange is a key ingredient in collaborative e-business protocols. BPML engages a message-based business process in which everything that interacts with a process does so through the exchange of messages. In Sterling B2B Integrator, business processes interact with services/adapters and other business processes.

Exchanging messages with services and adapters

In Sterling B2B Integrator, whenever a business process runs an adapter or service, there is an exchange of two messages. First, the business process sends a message to the service/adapter that contains the parameters that the service needs to run. The values of these parameters can either be constants, or they may come from process data. After a service is completed successfully, it sends its results back to the business process in a second message. The business process can select some of the contents of the message and place them into process data. Note that this message exchange occurs synchronously (the process must wait for the response message before continuing).

Exchanging messages between business processes

In Sterling B2B Integrator, messages are exchanged with business processes by way of the produce and consume BPML activities. The produce activity sends a message and the consume activity receives a message from a process.

When the produce activity runs, it sends a message to the specified consuming processes; the produce does not wait for the message to actually be consumed. If the receiving process for the message is not correctly running a consume activity, then the message is queued until a matching consume activity runs. The message exchange is thus considered asynchronous.

While BPML supports this message exchange, it is a best practice in Sterling B2B Integrator to use the Invoke Business process service to start a child or parallel process and pass message information.

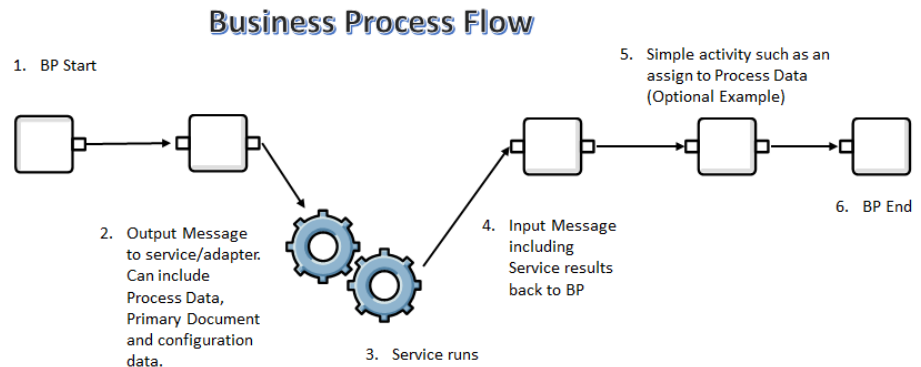
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What is BPML?

(Continued)

Message Exchange Example

This diagram shows a message exchange from the business process context to an adapter or service. While we usually think of the work being done from the point of view of the service, the message flow is always from the point of view of the BPML code being executed.



Sterling B2B Integrator BPML Activities

Overview

Activities perform BPML functions. BPML functions include operations that allow for the control of the flow of data through a business process. For example, the choice activity allows data to flow through the system that is based on the result of a decision.

BPML supports three main types of activities:

- Simple
- Complex
- Process

Remember!	Sterling B2B Integrator views Process activities as a kind of Simple activity.
------------------	--

Simple Activities

Use simple activities such as produce, consume, and assign to perform only one function. For example, the produce activity sends a message to another process. The assign activity is used only for setting a value.

Example:

```
<assign to="X">7</assign>
```

This assign command merely sets a value named X to 7.

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Sterling B2B Integrator BPML Activities (Continued)

Complex Activities	Use complex activities, such as sequence, all, and choice to model the flow of control. Complex activities can consist of one or more simple activities or another complex activity. These types of activities can be sequential, parallel, or conditional.
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Sequence Activity	A sequence activity runs all of the activities that are contained in it when a process is run. The activities are run in the order they are listed in the sequence.
--------------------------	---

Example BPML:

```
<process name="default">
  <sequence>
    <assign to="Recipient">joe@acme.com</assign>
    <spawn name=Send_Email"/>
  </sequence>
</process>
```

In this case, the sequence has two activities, Assign and Send E-mail. When the process is run, the system first runs Assign then runs Send E-mail because that is the order the activities are listed in the sequence.

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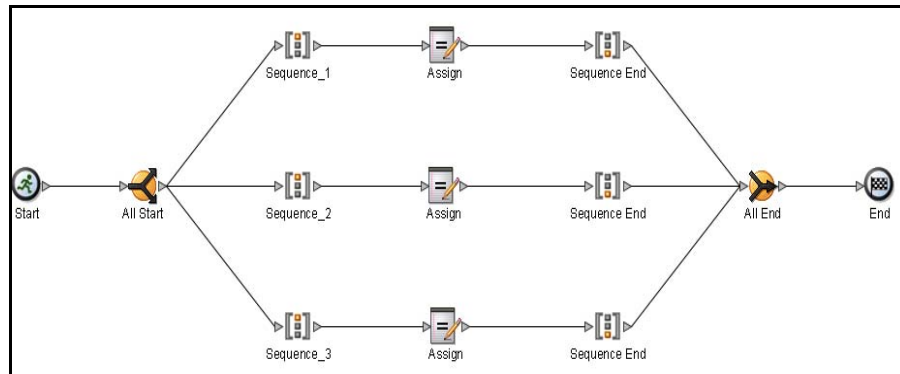
Sterling B2B Integrator BPML Activities (Continued)

The All Activity

The All activity is a complex activity that does not complete until all its sub, or child activities are run. In Sterling B2B Integrator, All runs all of its child activities simultaneously.

- Use the All activity in any position where an activity can appear.
- You must specify two or more sub activities.

The following figure illustrates a sample model.



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Sterling B2B Integrator BPML Activities (Continued)

The All Activity (Continued)

Example BPML:

```
<all>
  <sequence name="Subflow1_Start">
    <Subflow1>.....</Subflow1>
  </sequence>
  <sequence name="Subflow2_Start">
    <Subflow2>.....</Subflow2>
  </sequence>
  <sequence name="Subflow3_Start">
    <Subflow3>.....</Subflow3>
  </sequence>
</all>
```

This All activity has three child activities: Subflow1, Subflow2 and Subflow3. Sterling B2B Integrator attempts to run all three simultaneously. Since, running a main sequence corresponds to running children of the sequence, Sterling B2B Integrator performs all the activities in all three subflows simultaneously.

There is no requirement that the operations along the children of All move together in lockstep. However, nothing runs after the All until every activity in the All is completed.

Using the BPML All Activity Properly

The BPML activity, All (represented in the GPM by the All Start and All End icons), runs two or more complex child activities within a business process simultaneously. Because there are performance implications to using the All activity, use it only if you expect branches of your business process model to take significant amounts of time (in seconds or minutes) to complete (very few services take this long).

Do not use the All activity to run two or more instances of the same subprocess simultaneously. The system will handle the proper load balancing.

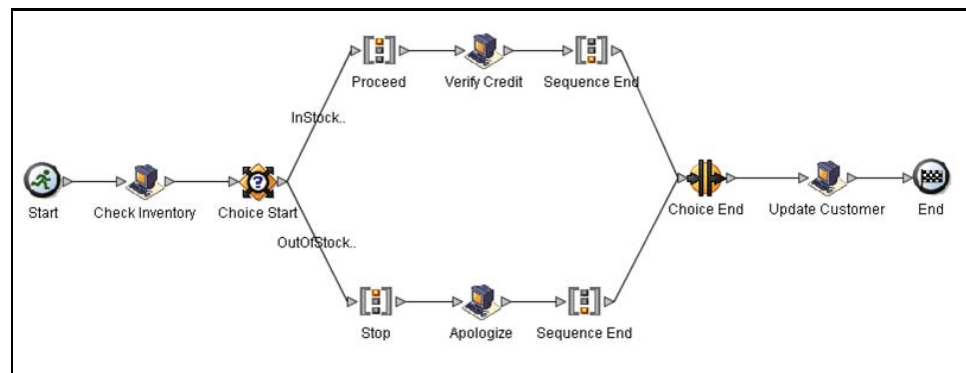
Sterling B2B Integrator BPML Activities (Continued)

The Choice Activity

The Choice activity is one of three types of complex activities. (The others are Sequence and All) A Choice activity is used to make decisions in the business process model and runs only one of the child activities it contains. The Choice activity makes it possible to model branch processing.

- Use the Choice activity any place where an activity can appear.
- Use a choice to model process branching.

The following figure illustrates an example model.



Process branching depends on a decision that is taken by the process and specifying which activity to run. One or more rules must be evaluated in order to make the decision. The select element is required and includes one or more case elements. Each case links the outcome of a rule to an activity. If the rule evaluates to true, or if the rule evaluates to false when the negative attribute is true, that activity is a candidate for execution.

Multiple cases can reference the same activity. However, the activity will run exactly once. If an activity is not referenced, it will never run. If it is determined that no activity is candidate for running, the choice completes immediately. Use a choice with a single activity to model a deferred activity. If two or more activities are candidates for running, only one is run.

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Sterling B2B Integrator BPML Activities

(Continued)

The Choice Activity (Continued)

In Sterling B2B Integrator, the order in which the case elements displays is the order in which they are evaluated. The first case which evaluates to true determines the child activity to run. The following figure shows an example BPML.

```
<process name="ProcessCustomerOrder">
  <rule name="BookInStock">
    <condition>foundBook = true </condition>
  </rule>
  <sequence>
    <operation name="Check Inventory">
      <participant name="InventoryService">
        <output message="checkStockRequest">
          <assign to='ISBN'>1-56592-488-6</assign>
        </output>
        <input message name="checkStockResponse">
          <assign to="foundBook" from="InStock" />
        </input>
      </operation>
    <choice>
      <select>
        <case ref="BookInStock" activity="proceed"/>
        <case ref="BookInStock"
          negative="true" activity="stop"/>
      </select>
      <sequence name="proceed">
        <operation name='Verify Credit Card'>
          </operation>
        </sequence>
      <sequence name="stop">
        <operation name='Apologize'> ... </operation>
      </sequence>
    </choice>
    <operation name='Update customer'> ... </operation>
  </sequence>
</process>
```

Focusing on the choice activity, it contains multiple edges, each of which has a rule. For now, think of a rule as something that is either true or false. In this case, before the process continues it checks to see whether the required book is in stock. The BookInStock rule is true if the book is in stock, otherwise it is false.

(Continued on next page)

Sterling B2B Integrator BPML Activities (Continued)

The Choice Activity (Continued)

Example BPML:

```
<rule name="BookInStock">  
  <condition>foundBook = true </condition>  
</rule>
```

When the choice is run, Sterling B2B Integrator checks each of the edges of the choice element and runs the path of the first rule it finds to be true. The Sterling B2B Integrator completes that activity, the choice itself is complete, and it runs no other activities in the choice after that point.

In this particular case, one of the choices results in an activity being run. It is possible; however to have choices where no activity is run because no rule was found to be true. An edge of a choice may have multiple rules; the process continues in a path if any of the rules in that path are found to be true.

In summary, the choice activity is used to make decisions in the business process. Rules that are associated with a choice conduct this decision making.

Activities Supported by BPML

Overview

This section provides a reference of the BPML activities that Sterling B2B Integrator recognizes. The activities are listed in alphabetical order for ease of reference.

All

The All activity is a complex activity that does not complete until all its sub activities are completed. Sub activities run in parallel.

Assign

The assign activity is a process activity that performs an assignment. You can use this activity any place where an activity can appear.

Choice

The choice activity is a complex activity that completes after one, or none of its sub activities completes. At most, one sub activity runs.

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Activities Supported by BPML

(Continued)

Overview (Continued)

Input

The input element accepts a message that is delivered to the process.

- Use the input element in the operation activity to accept a message that is delivered from a participant to the process (process input).
- Use the message attribute to reference the relevant message definition.
- Use the assign element to perform assignment from the message contents to the process data. You can use multiple assignments for multi-part messages.

Example:

```
<input message>  
  <assign/>  
</input>
```

Join

Use the join activity to make a process wait for the completion of the processes referred to by the **ref** attribute (an XPath expression for referencing an element in process data that holds process instance information).

Example:

```
<join ref>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

Overview (Continued)

OnFault

The onFault activity associates a fault handling the activity with a complex activity. Use it to recover from faults and allow process to continue. For more information about the onFault activity, see Using the onFault activity.

Operation

In Sterling B2B Integrator, the operation activity is used to call a service. Use it any place where an activity can appear.

- An operation involves a synchronous request/response message exchange with a possible fault message.
- When an operation is invoked, it delivers a request message and waits for a response message. If a fault is communicated, the operation faults.
- Use the operation activity to invoke an operation on a service.

The operation activity produces the outgoing message through assignment and waits for the incoming message to complete the activity. The participant element (or service) is mandatory. The output element must precede the input element.

Example:

```
<operation name="this is optional">
  <participant name="service name"/>
  <output message="what to send out">
    <assign />
  </output>
  <input message="what comes back">
    <assign />
  </input>
</operation>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

**Overview
(Continued)**

Output

The output element constructs a message that is delivered by the process to a participant. The message attribute references the relevant message definition.

- Use the assign element to construct the contents of the message by assignment from the process data.
- Use multiple assignments for multi-part messages, no assignments are required for empty messages.

Example:

```
<output message>  
  <assign/>  
</output>
```

Participant

The participant element defines or references a participant of the process. Sterling B2B Integrator recognizes this element only when it is used in the operation activity.

- Use the participant element to reference a participant within an activity.
- You must use the name attribute when defining the participant as a simple activity. Defining the participant as a simple activity defines a static participant.

Example:

```
<participant name>  
</participant>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

Overview (Continued)

Process

Use the process element to define a top-level process. The top-level activity relates to Start and End activities.

Example:

```
<process name>
  <rule name/>*
  <activity name/>
  / (simpleActivity | complexActivity)
</process>
```

Produce

Use the produce activity to send a message to a business process instance. The name attribute in the participant element must refer to an element in process data that provides the necessary information to contact a process instance.

As mentioned previously, it is a best practice to use the Invoke Service instead of Produce, Consume and Spawn BPML.

Example:

```
<produce name?>
  <participant name />
  (output message >
    <assign />*
  </output>
</produce>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

Overview (Continued)

Release Service

The Release Service is a system service that is used to remove extraneous data from the process data.

This service comes without predefined parameters, therefore you must use the Advanced Service Editor to create a parameter that called TARGET and assign a value to it. The value identifies the process data which is no longer needed.

TARGET is the name of the parameter the Release service expects in the "Message TO Service." The value that assigned to the parameter is an XPath expression which selects the nodes from the Process Data is to be released. Even though it is an XPath expression you do not need to check the "Use XPATH" check box.

Example:

If you put the following 'input' message into the Release service, it removes all the DOC-SPLIT tags from process Data.

```
<input>
<assign to="TARGET">/ProcessData/*[starts-with(name(),
'DOC-SPLIT')]</assign>
</input>
```

In this case, starts-with(name(), 'DOC-SPLIT') is an XPath expression that says, in effect, "Release all nodes in ProcessData with a name that starts with 'DOC-SPLIT'."

(Continued on next page)

(Continued)

Repeat

Use the repeat activity to run a portion of the business process multiple times. The **ref** attribute is the name of the activity to be run again. The referenced activity must be an ancestor of the repeat activity.

In most business processes the activity being repeated will be a sequence. The steps between the sequence start and end would be repeated if the sequence name is called by the repeat function. It is commonly used with a choice statement that when the choice criteria is met the repeat activity will be skipped and end the loop.

Example:

In this example the repeat function is named "Repeat" and the activity being repeated is a sequence name "ForEachDoc".

```
<repeat name="Repeat" ref="ForEachDoc"/>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

Overview (Continued)

Rule

The rule element defines a rule, the conditions by which the rule is met and the dependency on other rules.

- The condition element formulates an expression. The condition is met if the expression evaluated to true; or when the negative attribute is true, if the expression evaluates to false.
- Multiple conditions appearing inside a rule imply logical order. This notation is for convenience in reading. Other forms of logical composition are fully supported in the XPath language.
- The select element in a choice determines which line activity will run based on a number of cases. See the choice activity for more information.

Example:

```
<rule name?>
  <condition/>+
</rule>
<condition>
  expression
</condition>
<select>
  <case activity ref negative?/>*
</select>
```

(Continued on next page)

Activities Supported by BPML

(Continued)

Overview (Continued)

Sequence

The sequence activity is a complex activity that completes after all its sub activities are completed. All sub activities are expected in sequence. Use it any place where an activity can appear. A sequence models a compound state that transitions through a series of sub states, one for each sub activity. A transition is triggered upon the completion of a previous activity. This activity completes when the last sub activity is completed.

Example:

```
<sequence>
activity+
</sequence>
```

Spawn

Use the spawn activity to create a new instance of a business process. The ref attribute must contain the name of a valid business process within the Sterling B2B Integrator system. The new business process instance process data is a copy of the parent process data when the spawn activity ran. The child process and parent process are completed independent of each other; if one process completes (either normally or due to a fault), it has no effect on the other. After the spawn activity completes, an element in process data with the same name as the spawned process contains the instance information.

Example:

```
<spawn ref />
```

Process Rules

Introduction

A rule is an expression that evaluates single or multiple conditions that are based on information known to the process or information communicated to the process. The conditional expressions allow for diverse business process decisions both internal to the process and external to the environment.

Complex activities often require that a process select one of several alternative activities or even, on occasion, discriminate the information upon which it acts.

Rules can be built into activities that affect activity selection through branching and repeating actions and message consumption. In addition, rules can contain multiple conditions.

As the process acts upon rules, process branching may occur. Process branching is a result of a decision made by the process itself. This models the manner in which information created and collected through a process instance drives process running.

Rules express single or multiple conditions that are based on information known to the process or information communicated to the process. The contents of the condition activity must correspond to an expression that can be evaluated as either true or false. BPML mandates that these expressions be XPath expressions.

Rules for Choice activities

Complex activities often require that a process select one of several alternate activities or even, on occasion, discriminate the information upon which it acts. Rules can affect activity selection using the choice activity.

(Continued on next page)

Process Rules

(Continued)

Process Data

Process data is a context that exists for each process instance and can be used to hold or reference information that is accumulated during the life of the process. Process data is a private area that the process definition writer uses to store data that can be accessed by the running business process.

Data entering the system can be of many types, including a document and/or a set of name/value pairs and may be as complex and rich as XML itself. For example, when you run a business process from the User Interface, you can select an input file to use as a document. When a business process is started, it has data it puts into the document, and it can also set name/value pairs in the process data.

Important!	Services often use Process Data to store those name/value pairs that services down the line in the business might need.
-------------------	---

Using the Assign element

As a process runs, it obtains information from participants and communicates it to other participants. In Sterling B2B Integrator, participants can be services, adapters, or other processes.

An incoming document may contain a great deal of data, but the process participants may only need a small portion of that data. Therefore, the process needs to be selective about what it actually shares with different participants.

The process decides what information is relevant and uses the Assign element to share that subset. This takes a known value and includes it in an output message. In a similar manner, it can take a value from an input message and make it available for the process.

The direction in which an assignment is done depends on the context in which the Assign element appears; if it is part of the input, the assignment is from message to process; if it is part of the output, the assignment is from process to message.

(Continued on next page)

Process Rules

(Continued)

Using the Assign activity

The Assign activity sets something in the process data equal to a fixed value, either a number or a string; additionally, it can incorporate XPath expressions.

When using the assign element in an operation, you must know the parameters for the service. An example of this is the map name that the translator needs.

- The assignment source and target contexts depend on the manner in which the assign element is used.
- When the assign element appears within the output element, it performs assignment from the process data to the outgoing message. The To attribute corresponds to the path within the message and is used to construct the message contents.
- When the assign element appears within the input element, it performs assignments from the incoming message to the process data. The To attribute corresponds to a value within the process data.
- When the assign activity appears as an activity, it performs assignment from a constant to the process data. The To attribute corresponds to a path within the process data. The From attribute can be used to extract information from a previously assigned value.
- Sterling B2B Integrator does not recognize nested assigns.

(Continued on next page)

Process Rules

(Continued)

Using the Assign activity (Continued)

Example:

The schema for the assign activity as an independent activity is:

```
<assign to="x"> 3 </assign>
```

The schema for the assign element within an operation is:

```
<operation name="name">  
  <participant name="name of specific service"/>  
  <output message="output message from service"/>  
  <input message="input message for service">  
    <assign to='z' from ="x"/>  
  </input>  
</operation>
```

(Continued on next page)

Process Rules

(Continued)

Fault handling

A fault is an error that occurs while running any step of a process regardless of whether it is the result of a local error or an error in a service/adaptor. Fault handling allows the process to recover and proceed. A process may simply communicate the failure to its participants, or it could engage in recovery or select a different path to run. Generally, our main goal in setting up a fault handling procedure is to track any error and inform the operator.

Faults are propagated upwards through a process from the faulting activity to its parent activity. If the parent activity does not have an appropriate onFault to handle the error, it is propagated up to its parent, and so forth. If several activities are running in parallel, only the activity with the fault is affected. However, when the top-level activity fails, the process is aborted. Any activity associated with the process will complete, but the process will still abort.

Using the onFault activity

Errors are handled in BPML by the onFault activity. An onFault activity is associated with the complex activity it is contained in, and is only run if an activity in the complex activity results in an error.

- The fault handling activity runs if the activity faults before completing. The faulting activity completes if the fault handling activity completes successfully.
- Use multiple onFault activities to handle fault codes. A set of onFault activities must use unique fault codes and at most one activity may omit the code attribute.
- The onFault activity without a code runs only if no fault with a matching code is found in the correct complex activity.

(Continued on next page)

Process Rules

(Continued)

**Fault handling
(Continued)****Example:**

```
<onFault code?>  
  activity+  
</onFault>
```

Exceptions

An exception indicates a possible failure condition. A service may need to generate a workflow exception when a required input parameter is:

- Missing.
- Invalid.
- Disabled. For example, a map or a workflow definition is disabled.

Use the following syntax to construct an exception:

```
new WorkflowException (String errorText, int reasonCode)
```

Workflow exception reason codes are:

- `public final static int GENERAL_PARM_ERROR = 0;` (Default). Use this in situations that require, for example, missing properties files.
- `public final static int MANDATORY_PARM_MISSING = 1;`
- `public final static int INVALID_VALUE_FOR_PARM = 2;`
- `public final static int RESOURCE_DISABLED = 3;`
- `public final static int NO_DOCUMENT = 4`

The workflow engine places the error text string for the workflow exception into the status report.

Exercise 2–1 Viewing BPML Code

Scenario You need to view the BPML code of existing business process that declares multiple attributes.

Steps to view the BPML code

1. Click **Go!** in the Admin Console Home page to open the Graphical Process Modeler.
2. Click the **Open** icon  to open the business process.
3. Browse to the Exercise folder on the Desktop and select the **DeclareMultipleAttributes.bp** file in the local system.
4. Click **View** in the menu bar and **Source**. The BPML code is displayed in the workspace.

```
<process name="default">
  <sequence name="Sequence Start">
    <operation name="Assign">
      <participant name="AssignService"/>
      <output message="AssignServiceTypeInputMessage">
        <assign to="," from="*"></assign>
        <assign to="partnername">IBMPartner2</assign>
        <assign to="item">Ipad</assign>
        <assign to="description">Ipad with new features</assign>
        <assign to="price">40000</assign>
        <assign to="quantity">5</assign>
      </output>
      <input message="inmsg">
        <assign to="," from="*"></assign>
      </input>
    </operation>
  </sequence>
</process>
```

Element Editors

Overview

BPML activities are installed and configured automatically when you install the GPM. However, some BPML activities, services, and service types may require you to perform additional configuration in the GPM, depending on the task your process model needs to accomplish. You can locate these configuration parameters in one of five different element editor types: Property, Service, Edge, Produce, and Consume. Each element editor provides configuration parameters specific to an activity or a service.

Use this editor:	To configure:
Property	Activities that require additional information such as Choice, Assign, and Sequence.
Service	Services that require additional information before they can function.
Edge	Rules for the connections that run from a Choice Start activity to the next activity in the process model.
Produce	The message and name of the receiving participant or activity.
Consume	The message a business process should expect and the name of the participant sending that message.

Element editor allows for the configuration of the BPML operations and services that are placed in the business process.

Quiz

- | | |
|------------------|---|
| Questions | <ol style="list-style-type: none">1. How many branches within a Choice can be executed as a result of a case statement?<ol style="list-style-type: none">a. Oneb. Twoc. Threed. Unlimited2. Services and adapters are used to exchange messages between business processes.<ol style="list-style-type: none">a. Trueb. False3. The All activity does not complete until all the sequences are complete.<ol style="list-style-type: none">a. Trueb. False4. Which of these elements are available in Operations of services/adapters? Select all the correct answers.<ol style="list-style-type: none">a. Outputb. Participantc. Inputd. Message To Servicee. Message From Service |
|------------------|---|

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Lesson Review

Completed Objectives

This lesson was designed to enable you to:

- Define BPML
 - Discuss the activities that perform BPML functions
 - Differentiate between services and adapters as they relate to Sterling B2B Integrator
 - Explain how Sterling B2B Integrator uses BPML
-

Graphical Process Modeler (GPM)

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Background

Introduction	This lesson describes the elements that make up the Graphical Process Modeler (GPM).
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Lesson objectives	This lesson is designed to help you to: ■ Install the Graphical Process Modeler ■ Use the Graphical Process Modeler ■ Understand Stencil Elements ■ Understand Element Editors ■ Understand Basic BPML Operations
--------------------------	--

Graphical Process Modeler

Overview

To help you build process models that accurately define your business processes, Sterling B2B Integrator offers the Graphical Process Modeler (GPM). The GPM is a stand-alone graphical interface tool that is used in Sterling B2B Integrator to create and modify business processes. The GPM converts the graphical representation of business processes to well-formed BPML (source code) and saves you the effort of writing code.

Important!	The GPM recognizes BPML coded in a text editor. Therefore, you can open a bpml file in a text editor and view its graphical representation in the GPM.
-------------------	--

Downloading and Installing the Graphical Process Modeler

Java Web Start

Sterling B2B Integrator requires that the correct version of Java is installed on the client machine to open and work with the GPM. This is usually the same version as what is required for Windows server. Always confirm with IBM Knowledge Center the correct Java for your release.

System Requirements

Java Web Start requires approximately 5.5 MB of disk space on the client computer. The minimum recommended available disk space for Java Web Start and the GPM is 100 MB.

Exercise 4–1 Opening Graphical Process Modeler

Scenario	You are required to log in to the Graphical Process Modeler with a login Id that has permission to create and manage business processes.
-----------------	--

Instructions	You can follow the steps that are listed to open the Graphical Process Modeler. 1. Open the Admin Console Home page. 2. Click Go! under Tools to open the Graphical Process Modeler. 3. Click Run in the Warning-Security pop-up window. 4. Enter the User Id as admin and Password as password. 5. Click Ok. The Graphical Process Modeler opens.
---------------------	--

You can also open the Graphical Process Modeler with the following steps:

1. Click **Business Processes** under **Administration menu** in the **Admin Console Home** page.
2. Select the **Manager** sub menu option. The **Business Process Manager** page opens.
3. You can click **Go!** next to Run Graphical Process Modeler under the **Graphical Modeling**.
4. Click **Run** in the Warning-Security pop-up window.
5. Enter the User Id as **admin** and Password as **password**.
6. Click **Ok**. The Graphical Process Modeler opens.

Note: The system may prompt you with a warning to allow Java Web Start to run. Please allow it to run.

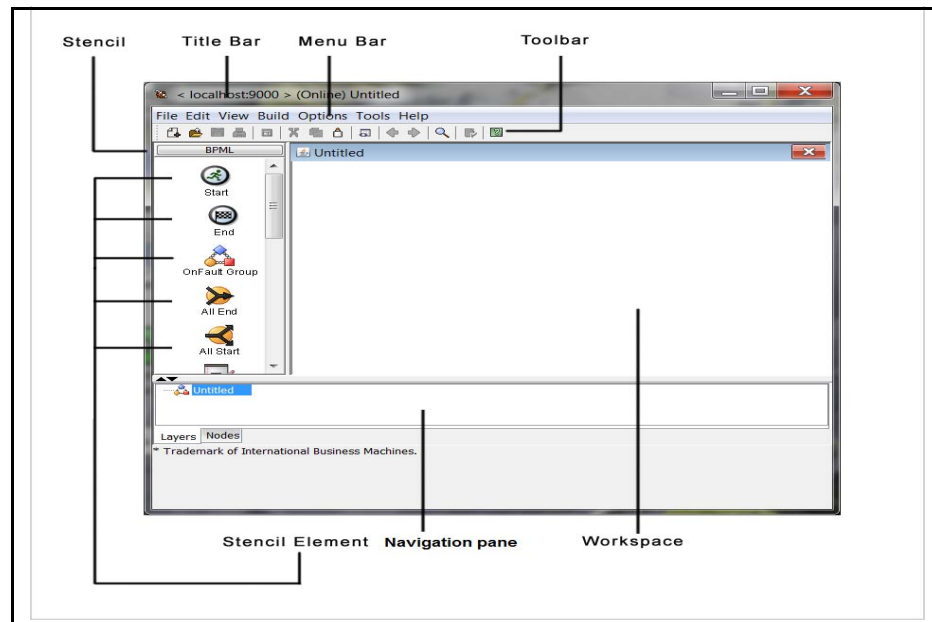
Note: If system prompts to restart after installation, close down any programs that are open and restart your computer. You can now start the GPM.

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Elements of the Graphical Process Modeler

Overview This section provides an overview of the major elements of the Graphical Process Modeler (GPM), such as menus, stencils, and workspace.

The following figure shows a standard GPM interface. A new business process model document is open, and the BPML and All Services stencils are visible.



(Continued on next page)

Instructor Note: Ask the students click File > New and follow along while you demonstrate.

Elements of the Graphical Process Modeler (Continued)

Anatomy of the Graphical Process Modeler

The following table lists the various elements of the Graphical Process Modeler along with their description.

Element	Description
The Menu bar	The menu bar provides access to the following menus: File, Edit, View, Build, Options, Tools, and Help.
The Tool bar	The tool bar contains shortcuts to the following functions: New, Open, Save, Print, Open Stencil, Cut, Copy, Paste, Source Manager, Previous, Next, Zoom, Properties, and Help.
The Stencils pane	The Stencils Pane provides the available stencils. To access different stencils, click the BPML Operations or All Services.
The workspace	The workspace shows the business process currently being built or edited.
The Navigation pane	The Navigation Pane shows a high-level organizational view of sub flows, activities, and services in the business process.
The Output window	The Output window shows BPML validation or error messages.
The Editor window	The Editor window shows the element editors, which are used to configure BPML activities and services.

(Continued on next page)

Elements of the Graphical Process Modeler (Continued)

The Menu Bar

The menu bar in the GPM is simple and complies with most Windows menu conventions. The following table describes the menu commands available in the GPM:

Menu	Description
File	The File menu allows you to: ■ Create a new business process model. ■ Open an existing business process model. ■ Close, save, and print a business process model. ■ Exit the GPM.
Edit	The Edit menu allows you to cut, copy, paste, delete, clear all, and select all elements from the workspace. The Edit menu also allows you to: ■ Add an annotation to an element. To add an annotation, you must first select an element icon. ■ Access element editors that contain configurable or non-configurable parameters. To view parameters, you must first select an element icon from the workspace.
Continued on next page	

Elements of the Graphical Process Modeler (Continued)

The Menu Bar (Continued)

Menu	Description
View	The View menu allows you to customize the GPM interface. In the View menu, click the option that you want to display or hide. When the option has a check mark next to it; then that menu option will be displayed. Options to display or hide are: ■ Source – Allows you to switch between the graphical representation and the BPML source code of a business process model on the workspace. The GPM does not allow you to edit or copy BPML source code. ■ Toolbar – Displays the bar containing shortcut command buttons. ■ Status Bar – Displays the bar at the bottom of the page. When you point to a menu option or tool bar button, the status bar displays a description of what that option does. ■ Output – Opens and closes the Output pane, which displays the error, validation, and system messages. This pane opens automatically when the GPM encounters an error during validation. ■ Navigation – Opens and closes the Navigation pane, which contains two tabs: Layers and Nodes. Layers provide a navigation structure in which you can determine the layer that you are currently working with. Nodes display all the nodes that exist on the layer that is currently shown on the workspace. For more information, see Navigation Pane on page 38. ■ Zoom – Allows you to change the magnification of a business process model. Zoom sets the magnification to 50%, 75%, 100%, 150%, or 200%.
Continued on next page	

Elements of the Graphical Process Modeler (Continued)

The Menu Bar (Continued)

Menu	Description
View	■ Grid – Displays grid markers or a ruler. The grid points allow you to align elements on the workspace. ■ Stencil – Shows the menu that lists stencils. Stencils contain labeled elements icons that represent BPML activities and services that you use to create business process models. From the Stencil menu, you can also close stencils as needed. ■ Refresh Services – Updates the GPM to include services and adapters that are recently configured. ■ Go To – Allows you to return to the previous flow view or move to the next flow view. ■ Auto Layout – Automatically aligns elements on the workspace.
Build	The Build menu command instructs Sterling B2B Integrator to validate a business process model against BPML syntax. After you complete the business process model, you must validate the business process before you can run it in Sterling B2B Integrator. The GPM might prompt you to perform a validation when you save the business process model. To disable or enable this prompt, select Preferences from the Options menu. Sterling B2B Integrator prompts you to validate a business process model during check-in; you can choose to delay the validation process.
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Elements of the Graphical Process Modeler (Continued)

The Menu Bar (Continued)

Menu	Description
Options	The Options menu allows you to specify GPM preferences. The Options are: ■ General – Allows you to specify whether the GPM should always validate on save or show the Validate on Save dialog box. ■ Source Manager – Allows you to specify Expert Mode, for the Source Manager. When you select Expert Mode, one dialog box with all the parameters needed to add a business process model to Sterling B2B Integrator. When Expert Mode is not selected, a series of dialog boxes appears for the parameters when adding a business process. ■ File Location – Allows you to specify a default directory to save all business process documents (.bp files) on your client computer. ■ Locale – Allows you to specify your location. ■ Service Editor – Allows you to choose override default service configuration values, which make the values that are defined in the Administrator UI as editable in the GPM. To specify your preferences, select Options > Preferences, and then select the appropriate tabs.
Continued on next page	

Elements of the Graphical Process Modeler (Continued)

The Menu Bar (Continued)

Menu	Description
Tools	The tools menu provides extended capabilities for building a business process model. Options are: ■ Rule Manager – Opens a dialog box that allows you to build and add XPath syntax to define business process rules and conditions. ■ Source Manager – Opens a dialog box that shows the business processes and versions of business processes that are available in Sterling B2B Integrator. Source Manager allows you to check business processes in to and out of Sterling B2B Integrator. ■ Add Sub Flow – Add a sub flow to a business process model. When adding a sub flow, the sub layers appears in the Navigation pane.
Help	The Help menu provides: ■ Connection to the Sterling Products web page ■ Help topics for using the GPM ■ Product version release and copyright information

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Elements of the Graphical Process Modeler (Continued)

- Stencils** The categorized stencils provide access to icons that represent services and adapters that are available in the GPM. These icons are used to create business processes that will run inside Sterling B2B Integrator.
- Using the labeled icons that represent services, you can indicate integration points and activities such as:
- BPML activities that control the flow of a business processes
 - Adapters that control the transmission of data between Sterling B2B Integrator and external applications
 - Services that carry out a specific activity (for example, translating file formats)
 - Customized services that are created and configured to meet a specific business need within the organization.

(Continued on next page)

Elements of the Graphical Process Modeler (Continued)

Stencils (Continued)

The Applications Stencil

The Applications stencil allows you to display stencil types that contain adapters that are installed to connect to external applications. You can display the following types of application stencils:

- ERP (Enterprise Resource Planning applications)

ERP applications adapters are:

- SAP® Suite adapter
- Oracle EBusiness Suite Adapter configuration service
- Oracle EBusiness Suite Message service

- IBM (IBM applications)

IBM applications adapters are:

- IBM® Sterling Connect:Direct® Server adapter
- IBM® Sterling Connect:Enterprise® UNIX Server adapter
- IBM® Sterling Information Broker
- IBM® Sterling Gentran adapter

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Elements of the Graphical Process Modeler (Continued)

Stencils (Continued)

BPML Stencil

Some BPML activities bracket other activities. That is, you must use these activities in pairs, at the beginning and end of the activities they enclose. When looking at BPML code they look like start and end tags. The following table lists the BPML icons with the description:

Icon	Description
 Start	The Start activity indicates the beginning of a business process. Every business process model must begin with Start.
 End	The End activity indicates the end of a business process. Every business process model must finish with End.
 OnFault Group	The OnFault Group activity is a sub flow that is used to associate a fault-handling activity with a complex activity. Use it to recover from faults and allow the process to continue.
 All Start	The All Start activity is a parent activity in a business process. All Start indicates that a business process is simultaneously executing all the child activities that are contained within the parent activity.
 All End	The All End activity indicates that all child activities within the parent activity are completed. Use All End to finish All Start.
 Choice Start	The Choice Start activity is a parent activity in a business process. Choice Start indicates a decision and runs one child activity as a result of that decision. You must assign a name to Choice Start using the Property editor and assign rules using the Rule Manager.
Continued on next page	

Elements of the Graphical Process Modeler (Continued)

Stencils (Continued)

Icon	Description
 Choice End	The Choice End activity indicates that the child activity that ran as a result of the parent activity is completed. Use Choice End to finish Choice Start.
 Sequence Start	The Sequence Start activity is a parent activity in a business process. Sequence Start runs a series of child activities in the order in which they are listed within the parent activity. When using Sequence Start, you must assign a name using the Property editor.
 Sequence End	The Sequence End activity indicates that the last child activity in a series of child activities in the parent activity is finished. Use Sequence End to finish Sequence Start.

(Continued on next page)

Elements of the Graphical Process Modeler (Continued)

Stencils (Continued)

The following table describes the BPML icons that allow extensive operations within a business process:

Icon	Description
 Assign	The Assign activity sets a value in the business process data that is equal to a fixed value. Inside an input or output element, the activity identifies a message that it should receive from a participant or identifies the contents of a message it should send to a participant. Configuration parameters allow you to indicate a fixed value or input/output message.
 Consume	The Consume activity reads a specific input message and stores that message in the business process data. Configuration parameters allow you to indicate a name for an instance of the Consume activity and the name of the message this activity should read.
 Produce	The Produce activity generates a message from the business process data and delivers it to a designated participant or activity. Configuration parameters allow you to indicate the message and name of the receiving participant or activity.

(Continued on next page)

Instructor Note: Inform students that the invoke service is commonly used instead of the Produce and Consumer activities.

Elements of the Graphical Process Modeler (Continued)

Stencils (Continued)

Icon	Description
 Spawn	The Spawn activity runs a nested business process. Configuration parameters allow you to indicate a name for an instance of the Spawn activity and the name of the activity the Spawn activity should run.
 Join	The Join activity merges the results of a completed nested business process into the business process data. Configuration parameters allow you to indicate a name for the business process for which the Join activity is waiting.
 Repeat	The Repeat activity runs a BPML operation one or more additional times. Configuration parameters allow you to indicate a name for an instance of the Repeat activity and the name of the activity the Repeat activity should run again.

For more information about activities, see Lesson 3, Basic BPML Concepts.

(Continued on next page)

Elements of the Graphical Process Modeler (Continued)

Services and Adapters	In the GPM, services and adapters are represented by labeled icons within stencils. For example, when you install the MQSeries v1.0 adapter, the MQ Series Adapter icon displays in the Messaging stencil. To add a service to a business process model, click and hold the mouse button on the service, and then drag it onto the workspace.
Title Bar	The title bar displays the file name of the business process model document currently open in the GPM.
Toolbar	The toolbar displays the buttons that provide quick access to commonly used commands. Unavailable buttons become available after you select a service on the workspace.
Workspace	The workspace is the area on the GPM interface to which you can drag icons to build a business process model. You can also view the BPML source code of a business process model in the workspace. To view BPML source code, from the View menu, select Source.

Exercise 4–2 Working with Stencils

Scenario	Use the following information to open and view the stencil elements and then close the stencils.
Instructions	For ease of understanding, the procedure is chunked into two parts as listed: ■ Opening a Stencil ■ Closing a Stencil
Opening a Stencil	You can use the following steps to open and view a stencil. 1. Open the Graphical Process Modeler. 2. Click File > New. 3. Click View > Stencil > All Services to display the All Services Stencil. 4. Click View > Stencil > BPML to display the BPML Stencil. 5. Click All Services to display Stencil elements.
Closing a Stencil	You can use the following steps to close a stencil. 1. Select Stencil from the View menu. 2. Complete one of the following steps: 3. Click Close Current Stencil to close the current stencil (topmost). 4. Click Close All Stencils to close all stencil.

(Continued on next page)

Element Editors

Overview	BPML activities and a few services are pre-configured when you install Sterling B2B Integrator. However, some BPML activities and services require you to perform additional configuration in the GPM, depending on the task your business process model must accomplish. You specify these configuration parameters in one of five different element editor types: Property, Service, Edge, Produce, and Consume. Each element editor provides configuration parameters specific to an activity or a service.
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Element Editors

(Continued)

Property Editor	The Property editor displays BPML icons that require additional information; such as choice start, assign, and consume icons.
------------------------	---

Using the Property editor, you can also define values to indicate errors. If you define values that indicate errors and the business processes encounters these values while running, Sterling B2B Integrator generates a report that describes the error.

Service Editor	The Service editor displays service parameters for each service and adapter. You can use the Service editor to:
-----------------------	---

- Perform additional configuration for services and adapters with active parameters.
- Apply values using the preset values in the GPM to define conditions for configuration parameters.
- Create your own name-value pairs, adding XPath expressions when needed, to define the path to name-value pairs that appear in business process data.

If your business process requires a service that is not available, contact your system administrator.

(Continued on next page)

Element Editors

(Continued)

**Service Editor
(Continued)**

The following table describes the parameters that display in the element editor for services and adapters. In addition to parameters specific to each service or adapter, you must configure these parameters

Field	Description
Name	Name of the type of service that you are using in your business process model.
Continued on next page	

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Element Editors

(Continued)

Service Editor (Continued)

Field	Description
Config	Service configuration to use in your business process model. Note: After you select a service configuration, additional configuration parameters display.
Output Msg	(Message to service) Sends one of the following types of information: ■ Message Only – Send only service configuration information to a service. ■ Obtain message First, then Process Data – Send service configuration information, name-value pair in the Service editor, and business process data. If duplicate information exists between the service configuration and business process data, the business process data overrides the service configuration information. ■ Obtain Process Data First, then Message – Send service configuration information, name-value pair in the Service editor, and business process data. If duplicate information exists between the service configuration and business process data, the service configuration information overrides the business process data. Note: When sending name-value pairs, you must indicate in the Service editor the name and value that describes the path for sending the name-value pairs.
Input Msg	(Message from a service) When a service sets variables in business process data: ■ Allow message write – Do not save the variables. ■ Allow process data write – Save the variables.

(Continued on next page)

Element Editors

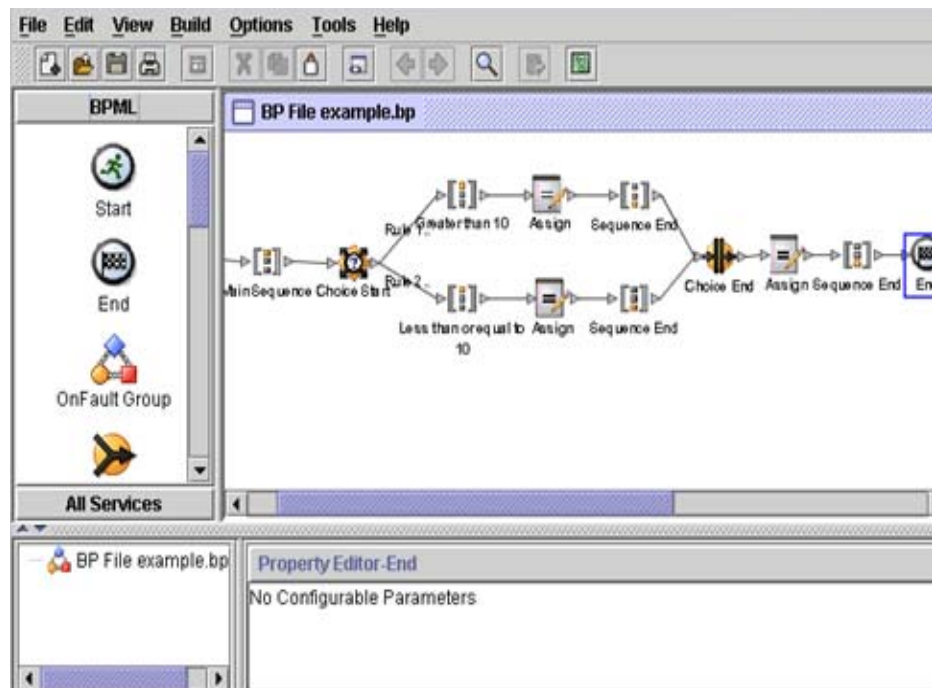
(Continued)

Edge Editor	The Edge editor lists rules that were created on the Rule Manager. You can assign these rules to links that run from a Choice Start icon to the next icons in the business process model.
Produce Editor	A Produce editor displays configuration parameters that allow you to indicate the message and name of the receiving participant or activity.
Consume Editor	The Consume editor allows you to indicate the message a business process should expect and the name of the participant sending that message.

Navigation Pane

Introduction

The Navigation pane contains two tabs: Layers and Nodes. When used together, the Layers and Nodes tabs allows you to quickly navigate through a business process model. The following figure shows the location of the Navigation pane:



(Continued on next page)

Navigation Pane

(Continued)

Layers	The Layers tab provides a relationship structure that shows each layer of a business process, including sub flows (nested layers). To expand the structure, click the root of the layer. When you expand the view, all the activities in the layer display on the workspace. A layer view is especially helpful when adding activities that handle processing errors within a business process.
Nodes	The Nodes tab provides an alphabetical structure displaying nodes (activities) that make up each layer in a business process. Select a layer from the Layers tab and then click the Node tab to view icons that make up the node in the Node tab and on the workspace. Selecting properties for an icon on the Node tab displays the configurable parameters for the node in the applicable element editor.

Quiz

- | | |
|------------------|--|
| Questions | <ol style="list-style-type: none">1. Which service editor option allows you to send service configuration information to another service?<ol style="list-style-type: none">a. Configb. Message From Servicec. Input Msgd. Output MSG2. Which of the following GPM elements opens the Edge Editor?<ol style="list-style-type: none">a. OnFaultb. Consumec. Spawnd. Line off a Choice Element3. You can open a bpml file in the GPM and view its graphical representation.<ol style="list-style-type: none">a. Trueb. False |
|------------------|--|
-

Lesson Review

Completed Objectives

This lesson was designed to allow you to:

- Install the Graphical Process Modeler
 - Use the Graphical Process Modeler
 - Understand Stencil Elements
 - Understand Element Editors
 - Understand Basic BPML Operations
-

Using the GPM

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Background

Introduction	This lesson provides instructions for using the Graphical Process Modeler (GPM) to create an executable business process.
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Lesson objectives	This lesson is designed to help you to: ■ Define a business process and the life cycle of the business process ■ Create a business process ■ Implement a business process ■ Check in a business process model ■ Enable a business process model ■ Manually run a business process model ■ Check out a business process ■ Modify a business process ■ Check in a modified business process ■ Disable a business process
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Business Processes

What is a Business Process?

Business processes are like road maps. Just as you use a road map to determine the best route through the countryside, Sterling B2B Integrator uses business processes to find the best route through critical business systems, such as purchasing, inventory status, and invoicing.

By analyzing and documenting your business and technical needs, you can determine your business processes. You can then reword those business processes to fit your business needs.

The Lifecycle of a Business Process

This section provides an overview of how Sterling B2B Integrator works. It follows Sterling B2B Integrator through:

- Analyzing a business process
- Defining a business process with services
- Compiling a business process
- Check In a business process
- Starting a business process
- Archiving a business process
- Restoring a business process
- Deleting a business process

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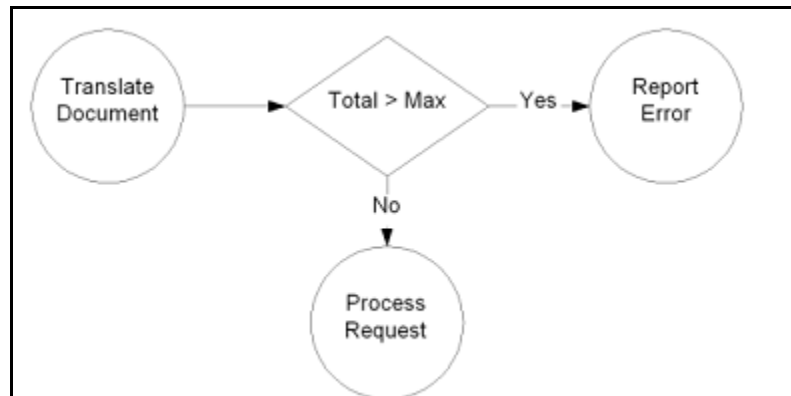
Business Processes

(Continued)

Defining a Business Process

Sterling B2B Integrator is a framework that runs customer-specific business processes. All operations that are carried out on documents and their order is set by an XML-based business process model.

Defining a business process might look like the following diagram. The circles represent activities and the diamond represents a decision point.



Since Sterling B2B Integrator stores business process definitions in XML, there are a couple of ways for a business process analyst to define them:

- Graphical Business Process Modeler included with Sterling B2B Integrator
- Simple text editor

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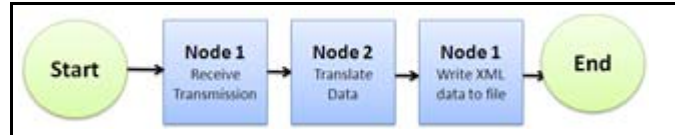
Business Processes

(Continued)

Defining a Business Process (Continued)

Nodes in a business process model

Nodes in a business process model are the graphical representation of the steps the business process goes through as it runs and processes the data involved. The following figure is a simple example.



Services

Flexibility is key to the business analysts who works with the IT staff to determine how business processes needs to be written. Sterling B2B Integrator allows you to write subprocesses with multiple reusable components or as one large service that does a complete process on its own. One way to look at services is to look at where they get the information they process and where they pass it when the processing is complete:

- Internal services are located inside Sterling B2B Integrator; they accept parameters and produce results, but they do not interact with any system outside Sterling B2B Integrator.
- Input services receive data from an outside system.
- Output services send data to an outside system.

(Continued on next page)

Business Processes

(Continued)

Defining a Business Process (Continued)

Adapters

Input and output services are known collectively as adapters. Adapters interact with the “world” outside Sterling B2B Integrator.

Business Process Context

- The service accepts a business process state and produces a modified business process state, or business process context (BPC). BPC also be referred as WFC, or Workflow Context.

Compiling a Business Process

After you create a business process model, you must validate and compile the process before you can run it. Validating ensures that all activities in the business process model are properly configured.

Compiling the business process allows Sterling B2B Integrator to determine the “Start” node ahead of time so business processes are more easily instantiated and run. Since the start pointer for the next activity (node) is stored in the current activity (node) information, compiling the business process also eliminates repeated parsing of XML and reduces the number of queries to the database.

Validating and compiling are done when you validate when saving or using the Validate button inside the GPM. Sterling B2B Integrator will not allow you check-in a non-validated business process.

When writing BPML or copy and pasting bpml text in the Administration interface you will have to validate before being able to move on during the check-in process.

Note: Because a business process is validated does not mean it will run without errors at run-time. It will only verify that the BPML is syntactically correct. It can not verify that the business process will interact with other systems or files correctly.

(Continued on next page)

Business Processes

(Continued)

Checking In a Business Process

Before you can use a business process within Sterling B2B Integrator, you must first check it into the system. You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface. This is the process of adding the Business Process from where you are developing it to the server where it will be executed.

Both the Sterling B2B Integrator interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time that you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

(Continued on next page)

Business Processes

(Continued)

Starting a Business Process

Business processes can be started either manually or automated by the system. Automated can be:

Scheduled- Set to run a specific times or intervals

Invoked- Called or started by another business process.

Bootstrapped- Similar to Invoked, but called from an adapter or event instead of a business process. Such as an adapter monitoring a file folder and kicking off a business process when a file is found.

Dynamic bootstrapping is the selection of the appropriate business process to run when data enters. Input data must enter Sterling B2B Integrator by way of an input adapter. This adapter takes the received data, puts the data and any metadata it finds into an Initial Business Process Context (IBPC), and begins bootstrapping. A business process definition is selected based on the data and/or metadata and instantiated as needed.

Archiving a Business Process

Archiving your business process data not only protects your critical data but also conserves database storage, improving the efficiency of Sterling B2B Integrator.

(Continued on next page)

Business Processes

(Continued)

Restoring a Business Process

After you have archived business process data to an offline location, you can restore that data and view it in Sterling B2B Integrator.

Deleting a Business Process

In Sterling B2B Integrator, you can delete individual business processes (except business processes used by Sterling B2B Integrator). Some reasons that you might delete a business process are:

- Change in Business activities and, as a result, a business process can no longer accomplish the activities.
- You need to remove versions of a business process to reduce the database size.

Note: A business process can not be deleted if there is still a record of an instance in the database. All instances of a bp must be purged from the database otherwise tracking could be impacted.

For example if a bp was executed and is set to not be purged for 3 days, that business process model could not be deleted until 4 days later. This is because there is still an instance record of that bp executing in the database for 3 days after it ran.

(Continued on next page)

Instructor Note: Students do this exercise, walk around, and help where appropriate. Make sure that all students have completed before moving on to connecting the icons in the next exercise.

Business Processes

(Continued)

Creating a Business Process

After analyzing your business needs, determining your business processes, and documenting the results, you must create business process models that represent those business processes. You can then implement and run the business processes with Sterling B2B Integrator.

Important!	Because a business process must be validated to run successfully in Sterling B2B Integrator, you must ensure that you validate the BPML code in each business process before you add the process to Sterling B2B Integrator. The sections in this lesson include steps to validate a business process
-------------------	---

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Instructor Note: Students do this exercise, walk around, and help where appropriate. Make sure that all students have completed before moving on to connecting the icons in the next exercise.

Business Processes

(Continued)

Implementing a Business Process Using the GPM

You can implement a business process in Sterling B2B Integrator with the Graphical Process Modeler (GPM) by following the appropriate steps here:

1. Create a business process model, including:
 - Configuring additional parameters
 - Adding sub flows
 - Assigning rules and conditions
 - Assigning name-value pairs that represent data to locate in process data
 - Configuring the Correlation service to generate correlations for business processes and document exchange
 2. Validate and save the business process model document.
 3. Add the business process to Sterling B2B Integrator.
-

Exercise 5–1 Moving Activities to the Workspace

Scenario You are required to create a business process to check for the quantity of Gizmo items from the inventory XML report. If the quantity is less than the minimum inventory level then the process should display a message to order more items and if the quantity is greater than minimum inventory level then the process should display a message not to order more items.

Instructions **Procedure to Move Activities to Workspace**

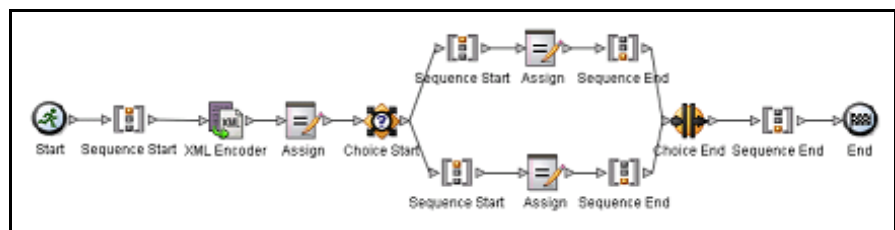
Follow the steps that are listed to move the activities to the workplace.

1. Open the Graphical Process Modeler.
2. Click New.
3. Bring up the Stencils if not already showing.
 - BPML Services
 - All Services
4. Move the following operations and services to your workspace by dragging the icons from the Stencil to the workspace:
 - Start (1)
 - End (1)
 - Sequence Start (3)
 - Sequence End (3)
 - Assign (3)
 - Choice Start (1)
 - Choice End (1)
 - XML Encoder (1)

Exercise 5–2 Connecting Activities

Scenario You need to connect the activities that determine the order in which the activities are run within the Business Process.

- Instructions** Follow the steps that are listed to connect the activities within the business process.
1. Click the triangle at the right of the **Start** node and keep the mouse button down.
 2. Drag the mouse to the left triangle of the **Sequence Start** node and release the mouse button. This creates a link between the nodes.
 3. Connect the activities from the previous exercise as shown in the following figure.



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Exercise 5–2 Connecting Activities

(Continued)

Instructions (Continued)

4. (Optional) Select **View > Autolayout** to visually align the business process model. Autolayout will evenly space all of the icons. This could move icons off your screen.
5. Select **Save** and save the business process model document (.bp file) to your client computer. You do not need to validate this business process model because you have not configured any parameters yet.

Remember!	Give the business process model a name that helps identify the business process.
------------------	--

Important!	If you do validate you will receive an error that ... is not configured or other errors. This is normal as you have not configured anything. You will configure the icons in later lessons.
-------------------	---

6. Select a directory on your client computer in the Save dialog box, and name your file **BasicInventoryProcess**, and click **Save**.

You are now ready to configure parameters, add sub flows as necessary, and assign rules and conditions for your business process model. These topics will be covered in the next lesson.

Important!	Certain activities and services require configuration within the GPM. You can perform the configuration with the five element editors in the GPM: Property, Service, Edge, Produce, and Consume.
-------------------	--

Viewing the Element Editor and Configuring Additional Parameters

Example Procedure to Configure Additional Parameters

These are the steps you would take to view an element editor and configure additional parameters.

1. Select an icon on the workspace and click **Edit > Properties** to display the applicable element editor. For example, when you select the Assign icon, the Property editor opens and provides parameters to configure for the Assign activity.
 2. Configure the parameters in the element editor.
 3. Configure the parameters in the applicable element editor for each remaining icon on the workspace as necessary.
 4. Right-click the **Editor** tab and select **Close Editor Tab**.
 5. Validate and save the business process model.
-

Sub-Flow

Subflows in the GPM

When you create a subflow, you create a layer in the business process model when viewed in the GPM. You can expand the view to work with the components of the subflow by using the Layers tab in the Navigation pane.

While a subflow layer provides more design space in the GPM, it does not create a separately-saved business process model. Nor does it create a child business process that is called from a parent.

Subflows are only available in the GPM and are not supported in the actual BPML code. If you were to look at the source code of a business process that contains a subflow you would only see an additional sequence tag. Also if you modify the BPML code manually and then later load in the GPM the subflow information will not show as a separate layer but as a sequence in the main layer.

Exercise 5–3 Adding Sub Flows

Scenario You are required to add a sub flow that creates a layer in a business process model where you can define more details about an activity in a business process. As you design the business process model, identify where sub flows are possible and design them.

Instructions You can follow the steps that are listed to add a sub flow:

1. Select **Add Sub Flow** from the Tools menu. In the Navigation pane, the business process model expands to list the sub flow.
2. Delete the **XML Encoder** and first **Assign** service from the business process by clicking the icon and pressing the delete key.
3. Insert the **sub flow** icon between the **Sequence Start** and the **Choice Start** icons within the business process model.
4. Link the icons.

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Exercise 5–3 Adding Sub Flows

(Continued)

Instructions (Continued)

5. Select the **sub flow** icon, right-click, and select **Properties** to display the Property editor to rename the sub flow. Type **SubFlow1** in the Value field. When you click out of the Property editor, the sub flow icon displays the new name.
 6. Click the **sub flow** in the Navigation pane to display a new workspace that includes unlinked Start and End icons.
 7. Create a business process model for the sub flow.
 - a. Add the XML Encoder and Assign icons within the subflow.
 - b. Arrange the icons.
 - c. Link each icon, including the Start and End icons.
 - d. Align the business process model.
 8. Click the parent business process in the Navigation pane (the one that the sub flow is part of) to return to the main process.
 9. Save the business process model.
-

Assigning Rules and Conditions

Introduction

To indicate a decision point in a business process model, you must:

- Assign a rule that describes the decision.
- Provide the conditions under which the decision is true or false.

When running a business process that processes XML documents, you must use XPath to describe rules and conditions for processing the XML document.

Specifically, you must create XPath expressions that:

- Identify particular elements (or nodes) and their positions in both the input and output documents.
- Identify particular elements in the input document for copying into an output document or for further processing.
- Define any constraints of name-value pairs for elements.
- Calculate values that depend on other values.

You can then assign rules that represent the decision to complete activities in the business process. The following table shows sample names and rules for simple syntax:

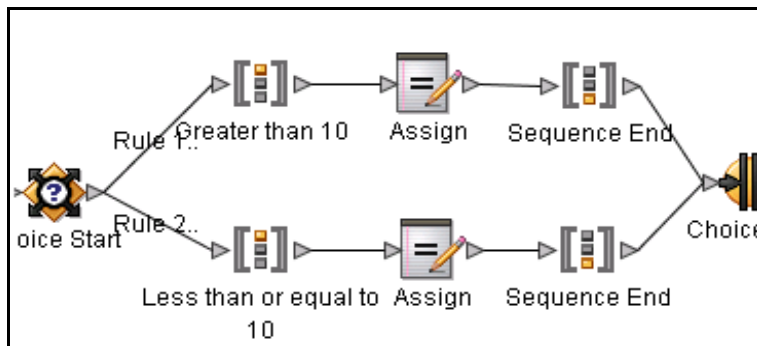
Name	Rule
rule1	X>10
rule2	X<=10
rule3	X<10

(Continued on next page)

Assigning Rules and Conditions

(Continued)**Introduction
(Continued)**

The following figure shows the part of a sample business process model where a decision must occur in a business process. The Choice Start and Choice End icons are included and rule names appear midway through edges (links) in the business process model.

**Rule Manager**

The Rule Manager is a tool in the GPM that is used to create rule conditions that will be evaluated in your business process. A rule is written in XPath and evaluates XML data in Process Data.

Edge Editor

The Edge Editor is used to assign the rules you created to a choice in your bp. Bring up the Edge Editor by double clicking a line coming off of your Choice Start. You must have a rule already created because the Edge Editor only allows you to select a rule and then say if it is true or false. This is why the name of the selected rule appears on the line.

(Continued on next page)

Assigning Rules and Conditions

(Continued)

Creating a Rule

You can create a rule that is defined by simple syntax or complex syntax. For ease of understanding, this procedure is chunked into two parts:

- Create a rule defined by simple syntax
- Create a rule defined by complex syntax using XPath

Create a Rule Defined by Simple Syntax

You can create a rule that is defined by simple syntax in the following manner.

1. Select **Rule Manager** from the Tools menu.
2. Click **Add** and type a name that describes the rule.
3. Type the syntax in the Expression field, and then click **OK**.
4. Repeat steps 1-3 until you finish creating rules that are defined by simple syntax.

(Continued on next page)

Assigning Rules and Conditions

(Continued)

Create a Rule Defined by Complex Syntax using XPath

You can follow the steps listed to create a rule defined by complex syntax using XPath.

1. Select **Rule Manager** from the Tools menu.
 2. Click **Add** and type a name that describes the rule.
 3. Click the button to the right of the Expression field to display the XPath Expression Builder.
 4. Display the expression type that you want to perform, double-click the function to add it to the Expression field, and then type values to complete the expression type as needed.
 5. Display the token set attribute to complete the function of the expression type and double-click the attribute to add it to the Expression field.
 6. Repeat steps 1-5 until you finish defining this XPath expression and click **OK**.
 7. Repeat steps 3-6 until you finish creating rules that are defined by complex XPath.
-

Exercise 5–4 Using Rules

Scenario	The rule manager allows for the definition of expressions that evaluate to true and false. The rules can be used for conditional execution of activities within the Business Process.
-----------------	---

Instructions	You can follow the steps that are listed to use a rule. 1. Click Tools > Rule Manager from the GPM main menu. 2. Click Add to bring up the Rule Editor. 3. Populate the Name box with Inventory Count. 4. Type the following string in the Expression box: <code>count(INVENTORY/PRODUCT) > 10</code> 5. Click OK in to add the rule to the list of rules in the Rule Manager. 6. Click OK on the Rule Manager window to save the rules.
---------------------	--

Exercise 5–5 Assigning a Rule

Scenario	You are required to assign rules and define conditions.
-----------------	---

-
- | | |
|---------------------|--|
| Instructions | <p>You can follow the listed steps to assign rules and define conditions.</p> <ol style="list-style-type: none">1. Click the edge (link) between the Choice Start and the Sequence Start icon on the top to display the Edge editor.2. Click Add to display the name and value columns in the Edge editor.3. Click in the field and select Inventory Count under the Name column.4. Select True for the selected rule under the Value column. When you click out of the Edge editor, you see the rule name on the edge.5. Click the edge (link) between Choice Start and the icon instance on the bottom to display the Edge editor.6. Click Add to display the name and value columns in the Edge editor.7. Click in the field and select Inventory Count under the Name column.8. Select Not true for the selected rule under the Value column. When you click out of the Edge editor, you see the rule name on the edge.9. Save the business process model. Do not Validate during the Save. |
|---------------------|--|
-

(Continued on next page)

Exercise 5–6 Viewing and Configuring the Different Element Properties

Scenario You are required to view and configure the different element properties.

- Instructions**
1. Double-click the first **Sequence Start** operation in the model to open the Property Editor and type the following text in the Value column for the Name field: **Main Sequence**.
 2. Click **Sub Flow** in the Navigation Pane.
 3. Click the **Assign** activity that immediately follows XMLEncoder in the subprocess and enter the following information:

Name	Value
Constant	N/A
To	TotalPrice

4. Click the **XMLEncoder** Service to open the Service Editor and complete the following:
 - Select **XMLEncoder** from the Config drop-down list box.
 - Select **Use existing XML document** from the drop-down list box in the Value column for the **mode** field.
 - Select **Yes** from the drop-down list box in the Value column for the **output_to_process_data** field.
5. Click **BasicInventoryProcess** from the Navigation Pane, so the BasicInventoryProcess BP is displayed on the workspace.
6. Click the **TOP** Sequence Start and type the following text in the Value column for the Name field: Greater than 10.

(Continued on next page)

Exercise 5–6 Viewing and Configuring the Different Element Properties (Continued)

**Instructions
(Continued)**

7. Click the **BOTTOM** Sequence Start and type the following text in the Value column for the Name field: Not Greater than 10.
8. Click the **Assign** activity (Greater than 10) and enter the following information:

Name	Value
Constant	Do Not Order More Items
To	Result

9. Click the **Assign** activity (Not Greater than 10) and enter the following information:

Name	Value
Constant	Order More Items
To	Result

Exercise 5–7 Saving and Validating a Business Process

Scenario Validating a Business Process ensures that all activities are completed properly and the BPML generated from the graphic is syntactically correct. The output window at the bottom of the Graphical Process Modeler shows any error messages. Saving without validating saves the current graph so that you can open it in the GPM later. Without validation, you cannot execute the Business Process.

Instructions You can follow the steps that are listed to save and validate a business process.

1. Select **File > Save**.
2. Click **Yes** when asked to validate the process model upon saving.
3. Complete the **Save** dialog:
 - a. Select the folder location.
 - b. In the FileName field, enter BasicInventoryProcess.
 - c. Select **OK**.
4. If, the process model is not valid the output window appears at the bottom of the Graphical Process Modeler with a display of the error messages. Correct the errors and try saving the Business Process again.

Implementing a Business Process

Introduction

After creating or modifying a business process model, you must validate, check in, and test it before running it in Sterling B2B Integrator. Use a test environment for checking in and testing your processes before you run them in your production environment.

The first time that you check in a business process, you create a business process definition; a uniquely named copy of the process model stored in Sterling B2B Integrator. The business process model that is stored in Sterling B2B Integrator is independent of any .bp file of the process that you have saved in a local directory.

At any time, you can modify an existing process model and check in a new version to Sterling B2B Integrator, or save the modified copy of the process in Sterling B2B Integrator as a new process altogether. Version management is an important part of business process maintenance.

Checking in a Business Process

You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface.

Both the Sterling B2B Integrator interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time that you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

The check in procedure differs depending on your goal and the application you are using.

(Continued on next page)

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in

During the check-in procedure, Sterling B2B Integrator prompts you to select various processing options for a business process model. These options are related to archiving, document storage and tracking, persistence levels and other instructions that are related to your business needs. The settings that you choose can affect system performance. Before you check in a business process model, have an idea of what the appropriate selections are for your installation.

The following table describes the options:

Field	Description
Process Type	Type a name that identifies the business process model. This is the business process definition name that is stored in Sterling B2B Integrator.
Description	Type a description that helps identify the business process that you are creating. This is stored in Sterling B2B Integrator with the business process definition you are checking in. If the process model is an updated version of a business process, you may want to use a meaningful version number or some other content to identify the version.
Continued on next page	

Implementing a Business Process (Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Set Queue	Sterling B2B Integrator allows you to set performance optimizations by queue, defining queue levels to allocate resources. Select the previously allocated queue level that you want for this business process model for processing.
Use BP Queuing	If selected, the bp is placed in a bp queue for processing. This is the default and recommended method. To run the bp in BP in Non-BP Queued mode, clear the check box. In BP Non-Queued mode, however, the Persistence Level for the BP is zero by default, so the process cannot be manually restarted or resumed. Also some services are unable to operate with zero persistence
Enable Transaction	Select this option to instruct Application to treat the entire process as a single transaction so that either all of the steps complete, or, in the event of an error, none of them do. When an error occurs, no data is committed; data returns to its pre-process state. Note: Enabling transactions applies only to services that support transaction mode. See the reference information for specific services to determine whether this option is supported. By default, this transaction mode is not enabled. Note: If transaction management is already built into the process model (the model includes Start and End Transaction services), do not select this option or the process will fail.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Commit All Steps When There Is An Error	Select this option to have the business process commit all work to the database, at the time an error is generated. With this option, you can either roll back or commit all work before the error in the onfault block is called. If you select this option, all steps before the error are committed to database and the business process keeps running on the onfault block path. If you do not select this option, all steps before the error will be rolled back and the onfault block path is called. By default, this option is not enabled. Note: This option is valid when the BPML is configured with an onfault block and Enable Transaction is selected
Category	You can optionally enter a category name to which this process model belongs, creating the category. The category does not affect processing; categories are for future product enhancements.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Persistence Level	Select the level of data to retain for generating a status report that describes each step that the business process completes. The levels are: ■ Full - Retain all data for this business process, including associated documents, activities, and all process data. ■ Step Status (Engine May Override) - The recommended persistence level for most process models because it helps optimize performance while providing a level of recovery suitable for most business needs. Persists process data and documents for service steps according to the persistence level supported by the service in the step. ■ BP Start Stop (Engine May Override) - Choose this level if you know that you do not need persisted status data for successful processing. Because service level settings override the selection, you may have additional data in a recovery scenario. * Persists process data and documents for service steps according to the persistence level supported by the service in the step. ■ System Default - Assume that for the data, configuration is already defined in Sterling B2B Integrator to retain data. Persists data according to the global persistence setting configured in the properties file. ■ Step Status Only - Choose this level if you need only status information for each step and your recovery needs do not depend on additional data being saved. ■ BP Start Stop Only - Choose this level if your recovery needs do not depend on data being saved for the processing steps.
Continued on next page	

Implementing a Business Process

(Continued)

**Determining
Processing Options
During Check-in
(Continued)**

Field	Description
	■ Zero - This level does not persist any business process data for recovery or process tracking. Choose this level only if you are certain that you will not need the data. Note: Zero persistence is available only if the process is running in Synchronous start mode. ■ Error Only - Choose this level if you do not need tracking data for your process when it is successful and your recovery does not depend on process step data being saved. ■ BP Start Stop Only (No Errors) - The first step of a business process is persisted in full, and last step is persisted in minimal. However, when the business process runs into an error which makes it stop running, no error step is persisted. The business process stays in ACTIVE state. If the process runs into an error, and there is an onfault step in the business process, the business process will complete. ■ Override None No IC - Does not retain any details while the business process is running or in the current process, and will not retain any details after the business process is run.
Event Reporting Level	Select the level of event reporting that you would like to retrieve for this business process when it runs. The levels are: ■ Full - Generate events for the business process, including the business process start and end time, start and end times for all services or adapters running as a result of this business processes, and any resulting errors and exceptions. ■ Minimal - Generate events for the business process, including the business process start and end time and any resulting errors and exceptions. ■ None - Do not generate any event reporting.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Recovery Level	Select the level of recovery for this business process if the business process should halt during execution. The levels are: ■ Auto Resume - Resume the business process at the point at which the business process halts. ■ Auto Restart - Restart the business process from the beginning. ■ Terminate - Terminate the business process. ■ Manual - Require you to resume or restart the business process manually.
Document Storage	Select the level of document storage for documents that process when the business process runs. The levels are: ■ Database - Store documents in the database. ■ File system - Store documents in the file system. ■ Archiving and purging from the file system may require special handling. ■ System Default - Store documents in the file system or database, according to how you configured archiving and purging in Sterling B2B Integrator. ■ Inherited - If this is a subprocess, use the same storage type indicated for the parent business process.
Continued on next page	

Implementing a Business Process (Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Enable BP Execution Setting	This option displays only when you are using a clustered Application environment. Select this option to display the available choices for specifying the node to run the process on. If you do not specify a node, the system uses any available node. To select a node: 1. In the Select Node field, select the node on which you want the process to run. 2. In the Execution Node Specific field, specify whether the process must run on the selected node or can use others. Options are: ■ Preferred Node - Run the process on the selected node unless the node is unavailable (inactive for some reason). ■ Mandatory Node - Always run the process on the selected node (if the node is unavailable for some reason, the process will fail). Note: The Execution Role Specific field is reserved for future use.
Deadline Setting	Deadline setting indicates how much time the process may take to complete. Select the appropriate setting. Settings are: ■ Do not set deadline - Allows you to run the business process without a deadline. ■ Complete by Deadline - Allows you to specify a deadline time, in hours and minutes, by which the business process must complete process once it starts. ■ First Notification: Hours and Minutes - Allows you to specify whether to receive notification before a business process deadline. ■ Second Notification: Hours and Minutes - Allows you to specify whether to receive another notification before a business process deadline.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Document Tracking	Enable or disable this selection. Select this option if you want the documents that are involved in this process model to be trackable according to the configured document tracking settings. Document tracking at the business process level is disabled by default, and this setting overrides the global document tracking setting. If you are not using Application document tracking features, leave the option disabled, to reserve system resources for other tasks.
Set OnFault Processing	Enable or disable this selection. Select this option to enable the on-fault activity that is specified in the process to immediately execute in the event of a system error.
Life Span	Use these options to: Indicate the length of time that data is pertaining to each instance of this business process model will remain in the system after the process is completed, to be available for monitoring, tracking, and reporting activities. Specify whether to archive or purge the process-related data when the life span expires. ■ Process Specific - Number of days or hours, or days and hours combined, that data for each instance of this business process model are permitted to remain in the system. ■ Expired Business Processes should be - Select either Archived or Purged for expired data. Archived data is stored in your file system. ■ System Default - Select this option if you want the data for instances of this business process model to expire and be archived after two days. This option is configurable.

Implementing a Business Process

(Continued)

Enabling Document Tracking

Enabling Document Tracking makes the document activity for this business process available from the Advanced Search menu. Access Document Tracking from **Business Process > Monitor > Advanced Search > Documents**.

Setting onFault Processing

The onFault element is used to handle errors. You can include onFault elements in any complex activity for which it may be necessary to recover from faults so that the process can continue. The onFault element contains activities for handling faults.

Setting the Persistence Level

The persistence level options allow you to specify, at the business process level, the types of data to retain in the database for the process whenever it runs. Persisted data can be crucial to process recovery in the event of errors and is useful for monitoring and tracking activities, including generating status reports. Persisted data includes any of the following types of data. Status information for the steps in the process is composed of:

- General status information for the steps (start and stop time, step name)
- Error information for failed steps
- Data describing the start and stop conditions of the process
- Process data
- Documents that are associated with the process

The types of data persisted for a process depends on the activities the process involves (such as whether the process has documents that are associated with it) and your persistence level settings.

(Continued on next page)

Implementing a Business Process

(Continued)

Setting the Persistence Level (Continued)

Persistence can also be set globally (at the system level) for all processes; by default it is set to FULL persistence, although it may have been customized for your installation. If your installation uses the default FULL setting for the system, you generally do not need to change persistence at the business process level unless you are seeking performance improvements. For the process model you are checking in, your selection here overrides the global persistence level that is set in the properties file.

Important!	Two exceptions might affect your persistence selection. First, remember that some services support only a limited selection of persistence settings, regardless of the level of persistence you assign to the process model. Second, any persistence overrides configured in the BPML for steps in the process model override the setting that you make here for the specified step or steps.
-------------------	---

Choose the option that provides the type of data you need for monitoring, tracking, and recovery in the event of a failure. Remember that persisting data can slow system processing by filling up the database. Consider your archive and purge settings in addition to the amount and type of data you intend to persist.

(Continued on next page)

Implementing a Business Process

(Continued)

Choosing Document Storage

Document storage type is the storage type that you want for documents that process when the business process runs. The levels are:

- **Database** - Store documents in the database.
- **File system** - Store documents in the file system. This option can be best for large documents. Using this method, you must handle archiving and purging the documents from your file system.
- **System Default** - Store documents in the file system or database, according to how you configured archiving and purging in Sterling B2B Integrator.
- **Inherited** - If it is a subprocess, use the same storage type indicated for the parent business process.

When you check in a business process, Sterling B2B Integrator gives you the opportunity to select the location for storing the business process.

Setting the Process Instance Life-Span

When you check in a business process, Sterling B2B Integrator gives you the opportunity to set the Life Span of specific instances generated by that business process. The Life Span setting tells Sterling B2B Integrator how long the instance should remain in the active system. In addition, when the business process instances expire, you can set them to either automatically archive or purge themselves from the system.

(Continued on next page)

Implementing a Business Process

(Continued)

Setting the Expired Characteristics of a Process Instance

When you check in a business process, Sterling B2B Integrator gives you the opportunity to determine whether the specific instance of the business process needs to be Archived or Purged. This must be set up in the business process. Archive moves instances of business processes that have been run and any associated data from the live system to a storage area. Once in the storage area the business process instance (and data) is available to be moved to offline media for storage. Purge permanently removes business process instances that have run and any associated data. Archive and purge help to limit the amount of information that is kept in the live system. It is important to note, data that must be kept long term should be archived and stored in a safe media.

Enabling a Business Process

You must enable a business process that is checked in to the system before it can be available to run. Enabling a process model tells the system that this process is available to run. You cannot enable a business process unless it was validated at check-in. Business processes checked in through the Administration UI are automatically validated.

(Continued on next page)

Implementing a Business Process

(Continued)

Creating Permissions

Select this option to have a permission selection created for this business process model automatically. The permission can be assigned to user accounts to allow or restrict user access to running the process. Permissions are optional. If none are assigned, the process is available for all users.

Setting the Default Version of a Business Process

When you check a process model into the system, you can also set it as the default process. The default process is the process version that Sterling B2B Integrator uses when running a request of a new process instance that does not have explicit version information. The detail information about the versions of business processes is discussed in the later part of this lesson.

Disabling a Business Process

Only business processes set as the default version can run. A business process that was checked in to the system can be disabled, or made not available for execution, if it is not set as the default version. When a business process version is disabled, the system will not start new instances of the process.

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Implementing a Business Process

(Continued)

Check in the Business Process

You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface.

Both the Administration Interface or User Interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time when you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

There are three ways to check in a business process. The Business Process Manager provides two different methods. You can check in a business process by using either the:

- Graphical Process Modeler
 - Business Process Manager
-

Exercise 5–8 Check in a Business Process using Graphical Process Modeler

Scenario You are required to check in a business process using GPM.

Instructions You can follow the steps that are listed to check in a business process with GPM.

Important!	If you are not already in the Graphical Process Modeler, complete steps 1 and 2. Otherwise, skip to step 3.
-------------------	---

1. Click **Business Process > Manager** on the Sterling B2B Integrator Administration menu bar.
2. Click Go! on the Sterling B2B Integrator Business Process Manager page, and click next to **Graphical Modeling: Run Graphical Process Modeler**.
3. Click **Tools > Source Manager** in the Graphical Process Modeler.
4. Log on to the Source Manager with the Sterling B2B Integrator login and password.
5. Click **Process > Add**.
6. Type the Process Name **BasicInventoryProcess** and click **Next**.

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Exercise 5–8 Check in a Business Process using Graphical Process Modeler (Continued)

Instructions (Continued)

7. Browse out and select your BP file using the **Browse** Button.
8. Type **Basic Inventory BP** as the description. Click **Next**.

Important!	For the following fields, your view may differ, depending on whether you specified Expert Mode for the Source Manager through the GPM. If you did not specify Expert Mode, click Next to display the following fields.
-------------------	--

9. Set the Queue to **4**, the Persistence Level to **System Default**, the Event Reporting Level to **None** and the Recovery Level to **Manual**. Set the Document Storage to **System Default** and click **Next**.
10. Check the option, **Do Not Set Deadline**.
11. Accept the System Defaults for Archive on the next two screens by clicking **Next**.
12. Check **Validate**.
13. Check **Enable Process**.
14. Do not check **Enable Document Tracking** and **Set onFault Processing**.
15. Click **Finish**.
16. Click **OK**.

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Exercise 5–9 Checking in a Business Process using Business Process Manager

Scenario You are required to check in a business process using the business process manager.

- Instructions** You can follow the steps that are listed to check in a business process using the business process manager:
1. Go to **Business Process > Manager** from the Sterling B2B Integrator Administration menu.
 2. Go to **Create Process Definition** and click **Go!**
 3. Type the Process Name **BasicInventoryProcess1**.
 4. Click the radio button for **Check in Business Process created by the graphical modeling tool** under Select an input mode for defining the new process and click **Next**.
 5. Click **Browse** to locate the bp file on the Process Check-in screen and then click **Open**.
 6. Type a **description**, and then click **Next**.
 7. Uncheck **Document Tracking** and **onfault processing** on the Process Levels screen, set the Priority Level to **4**, the Persistence Level to **System Default**, the Recovery Level to **Manual**, the Event Reporting Level to **None** and the Document Storage to **System Default**. Click **Next**.
 8. Leave the amount of time set to **System Default** on the Life-Span Screen.
 9. Click **Next**.
 10. Click **Finish** on the Process Confirm screen.

Important!	The Enable Business Process checkbox is checked.
-------------------	--

Exercise 5–10 Run a Business Process Manually

Scenario Sterling B2B Integrator runs most business processes automatically. There are times however; when it is necessary to run a business process manually. This exercise describes the procedure for manually running a business process. When you build a new business process, it is important to test it before moving it into production. You can do this by manually running the business process.

Instructions You can follow the steps that are listed to manually run a business process.

1. Click **Business Process > Manager**.

The Business Process Manager screen will be appears.



(Continued on next page)

Exercise 5–10 Run a Business Process Manually (Continued)

**Instructions
(Continued)**

2. Type **BasicInventoryProcess** in the Search text box and click **Go!**
 3. Click the **Execution Manager** icon for BasicInventoryProcess.
 4. Click the **execute** icon.
 5. Click **Browse** and go to **Class Labfiles** to locate the test data. Your instructor will provide the details of the location of the lab files.
 6. Locate the file **InventoryOf3.xml** in the Class labfiles located in the Datafiles folder on the Desktop, and click **Open**.
 7. Click **Go!** Observe the separate Execute Business Process window while the business process runs.
 8. Close the **Execute Business Process** window when the Business Process has completed running.
 9. Repeat steps 4-9 using **InventoryOf11.xml** as the input document.
 10. Click **Return**.
-

Exercise 5–11 Obtaining Results

Scenario Follow along to see the record of your executed business process instance data. We'll learn more about tracking business process instance data later in this course.

Instructions Follow the steps that are listed to view the record of your executed business process instance data.

1. Click **Business Process > Monitor > Current Processes**.

The Business Process Monitor screen displays the current processes.



The screenshot shows the IBM Sterling B2B Integrator interface. The top navigation bar includes links for Admin Console, Channels, Operator, Community Management, and Advanced File Transfer. The left sidebar contains an Administration Menu with options like Business Processes, Monitor, Advanced Search, Central Search, Current Documents, Current Activities, Message Entry Workstation, Trading Partner, Deployment, Invoicing, EBICS, and Operations. The main content area is titled 'Monitor' and features a checkbox for 'Automatically refresh every minute'. Below this is a table with columns: Status, ID, Name, State, Started, Ended, Deadline, Parent/Child, and Ex. The table displays two rows of data for 'BasicInventoryProcess' with status 'Completed'. The last update timestamp is '03/02/2011 10:19:28 AM'.

Status	ID	Name	State	Started	Ended	Deadline	Parent/Child	Ex
Completed	12656	BasicInventoryProcess	Completed	3/04/05 12:26:52 PM	3/04/05 12:26:53 PM	None	None	
Completed	12650	BasicInventoryProcess	Completed	3/04/05 12:25:19 PM	3/04/05 12:25:20 PM	None	None	

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Exercise 5–11 Obtaining Results

(Continued)

Instructions (Continued)

2. Locate first instance of **BasicInventoryProcess** in the list.
 3. Click corresponding **Business Process ID** to view the Business Process Detail page.
 4. Click the **Info** icon in the Document column of step 2, the Assign Service.
 5. Click the **Info** icon in the Instance Data column of step 3.
 6. Click the **Info** icon in the Instance Data column of step 4.
 7. Click **Return**.
 8. Repeat steps 2-6 with second instance of **BasicInventoryProcess**.
 9. Click **Return** when complete.
-

Modifying a Business Process

Introduction

To modify or reuse a business process that is currently checked in to Sterling B2B Integrator:

1. Check out the business process.

The check-out process allows you to save the business process to your client computer. You can then open the checked-out business process file in the GPM and modify the business process model.

2. Modify or Edit the business process as needed.
3. Check the new version in to Sterling B2B Integrator when finished.

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Modifying a Business Process

(Continued)

Checking Out a Business Process

When you need to modify or reuse a business process that is currently checked in to Sterling B2B Integrator and enabled for modification, you can check out the business process from Sterling B2B Integrator. The check-out process allows you to save the business process to your client computer. You can then modify the business process document to create new versions of a business process or a new business process. If you have logged in to Sterling B2B Integrator using Source Manager in the GPM, you can check out a business process using Source Manager. Source Manager also allows you to check out one or many business processes at the same time.

You may want to check out a business process when:

- A business analyst creates or modifies the business processes and has limited access to the host.
- An employee modifies the business processes from a remote location.
- A business process exists on a test environment and is ready for use in the production environment.
- Sterling B2B Integrator checks out a copy of the selected process version, not the original version.

If the selected business process is enabled, it remains enabled until it is replaced by a newer version.

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Modifying a Business Process

(Continued)

Making Changes to an Existing Business Process Model

To change a process model that was checked into Sterling B2B Integrator, you first need to check out the process. This locks the process so that no other users will make simultaneous changes. The checked out version of a process model remains locked to other users until that user checks in the process.

The GPM is the preferred method to change a business process model, although advanced users prefer to change using BPML. When you choose to work in the BPML, and then reopen the process model in the GPM, it will open the process using the default layout options. Since the default layout can appear differently than your specified layout, it is recommend to work in the process modeler.

Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler

Instructions

You can use the following procedure to edit the particular business process.

1. Click **Tools > Source Manager** in the Sterling B2B Integrator Graphical Process Modeler.
2. Log on to the Source Manager with your Sterling B2B Integrator login and password.
3. Scroll through the list of business processes until you find **BasicInventoryProcess**.
4. Select **BasicInventoryProcess** process and click the **Check out a Process** icon.
5. Click **OK** to accept the location where BasicInventoryProcess.bp will be saved.

Important!	The Lock check box indicates that a user is operating on the business process.
-------------------	--

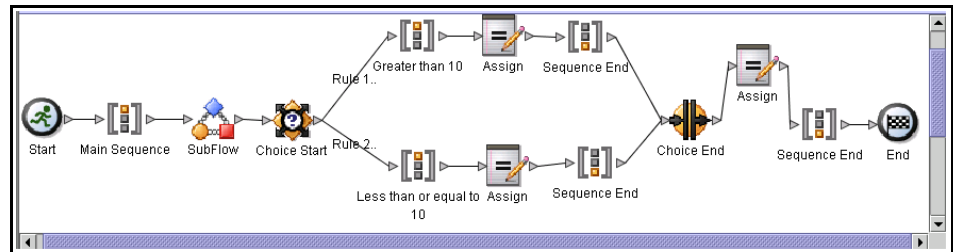
6. Click **Open file** icon in the Sterling B2B Integrator Graphical Process Modeler, and select the location where **BasicInventoryProcess** was saved.

(Continued on next page)

Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler (Continued)

Instructions (Continued)

7. Modify the business process model by adding **Assign** BPML activity.
 - Click and drag Assign to the workspace from the BPML Operations stencil.
 - Break the link after Choice End and before Sequence end.
 - Connect the new Assign to Choice End and the Sequence End respectively.



8. Right-click the **Assign** activity, then click **Properties**.
9. In the Property Editor:
 - Type **count(INVENTORY/PRODUCT)** in the **from** field
 - Type **TotalCount** in the **to** field

Important!

The **to** field cannot have any spaces.

10. Click **File > Save**.

(Continued on next page)

Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler (Continued)

**Instructions
(Continued)**

11. Click **Yes** on the Validate on save window.
12. Check in the process model.
13. Run the process model with either Inventory documents.

Important!	Be certain to validate again and enable during check-in; otherwise you cannot run the process.
-------------------	--

14. Observe **Info** icon of the Instance Data column of step 4.
-

Exercise 5–13 Editing a Business Process Using BPML

Instructions

You can use the following procedure to edit a particular business process.

1. Click **Business Process > Manager** on the Sterling B2B Integrator Administration page.
2. Type **BasicInventoryProcess** in the Search Process Name field.
3. Click **Go!**
4. Click the **Source Manager** icon for the BasicInventoryProcess business process definition.
5. Click the **Edit** icon for the version of business model you are planning to edit.
6. Type a **Description** for versioning.
7. Change the following line:

```
<condition>count (INVENTORY/PRODUCT) &gt; 10</condition>
```

to the following:

```
<condition>count (INVENTORY/PRODUCT) &gt; 15</condition>
```

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Exercise 5–13 Editing a Business Process Using BPML

(Continued)

8. Click **Validate** and click **Next**.
 9. Accept the Process Levels defaults. Click **Next**.
 10. Accept the default settings on the Deadline settings screen. Click **Next**.
 11. Accept the default settings on the Life Span screen. Click **Next**.
 12. Select the current version as the **Default**. Click **Next**.
 13. Click **Finish**.
 14. Run the process model with **InventoryOf11.xml** document.
 15. Observe the process. It now instructs you to “Order More Items”.
-

Exercise 5–14 Checking out the Business Process with Edited BPML

- | | |
|---------------------|---|
| Instructions | <hr/> <ol style="list-style-type: none">1. Select the BasicInventoryProcess in Source Manager, and select Process > Check out.2. Go to your working folder on your local system when prompted to save the process to your client computer, and click Save.3. Return to the GPM when the save is complete, and select File > Open.<ul style="list-style-type: none">■ Go to your working folder , and choose BasicInventoryProcess. Click Open. Notice the changes to the names of the Sequence Starts.4. Click Tools > Rule Manager.5. Click your rule, and select Edit. <p>Note that your rule now test for 15 instead of 10.</p> <hr/> |
|---------------------|---|

Version Control

Versioning

Sterling B2B Integrator maintains a record of each business process, when it is checked in, when it is checked out, and when it is modified. This simple process keeps two users from inadvertently overwriting each other's modified processes.

As each business process definition is checked into the system, a versioning record begins. Furthermore, as a different version of a process is checked in, you specify the default version of that process. This is how Sterling B2B Integrator knows which version of a process to run when new data enters the system or when a specific version is not explicitly requested.

Important!	You must also validate and enable the business process upon check-in for that version to become the default.
-------------------	--

The Execution Manager allows you to manually select which version of a business process to execute. This is useful when you are building a new version of business process and want to compare results. In addition, Sterling B2B Integrator can then keep track of which version of a process was running for a particular business process instance.

Important!	It is recommended that you use the description for setting the versions.
-------------------	--

Exercise 5–15 Viewing a Business Process Definition Version from within the GPM Source Manager

- | | |
|---------------------|---|
| Instructions | <hr/> <p>Use the following procedure to examine the version information for a particular business process.</p> <ol style="list-style-type: none">1. Click Tools > Source Manager in the Sterling B2B Integrator Graphical Process Modeler.2. Log on to the Source Manager.3. Scroll through the list of business processes until you find BasicInventoryProcess.4. Expand the view of BasicInventoryProcess to see the list of versions for this business process definition.5. Close the Source Manager. <hr/> |
|---------------------|---|

Exercise 5–16 Viewing a Business Process Definition Version from within Sterling B2B Integrator

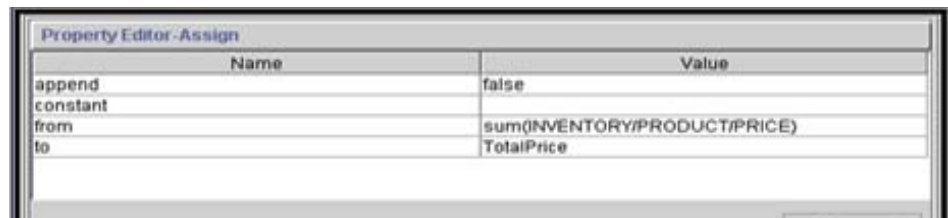
Instructions	Use the following procedure to examine the version information for a particular business process. 1. Click Business Process > Manager on the Sterling B2B Integrator Administration home page. 2. Type BasicInventoryProcess in the Search Process Name field. 3. Click Go! 4. Click the Source Manager icon for the BasicInventoryProcess business process definition. 5. Observe the version information in the Source Manager; specifically the Date and Change History columns.
---------------------	---

Exercise 5–17 Completing the Process

Instructions

In this exercise, an XPath expression needs to be added in the sub flow within the BasicInventoryProcess bp .

1. Open the GPM, if not already open.
2. Click **Tools > Source Manager** in the Sterling B2B Integrator Graphical Process Modeler.
3. Log on to the Source Manager with your Sterling B2B Integrator login and password.
4. Scroll through the list of business processes until you find **BasicInventoryProcess**.
5. Select **BasicInventoryProcess** process and click the **Check out a Process** icon.
6. Click **OK** to accept the location for where BasicInventoryProcess.bp was saved.
7. Click **Open file** icon in the GPM and select the location of where BasicInventoryProcess.bp was saved.
8. Click the **Sub Flow** in the Navigation Pane. Click the **Assign** activity in the model to open the Property Editor for the operation.
9. Remove the text in the Value column for the Constant field.
10. Type the following XPath expression text in the Value column for the From field: `sum (INVENTORY/PRODUCT/PRICE)`.



Name	Value
append	false
constant	
from	sum(INVENTORY/PRODUCT/PRICE)
to	TotalPrice

(Continued on next page)

Exercise 5–17 Completing the Process (Continued)

Instructions (Continued)

11. Click **File > Save**.
12. Click **Yes** on the Validate on save window. You should receive a success message where your editors have displayed.
13. Check in the process model.
14. Run the process model with either Inventory documents.
15. Obtain the results from running the business process.

If you encounter any errors or have questions on this exercise, contact your instructor.

Important!	Be certain to validate again and enable during check-in, otherwise you cannot run the process.
-------------------	--

Quiz

Questions

1. Which of the following statements are true regarding subflows? Select all correct answers.
 - a. An internal service is used to call other business processes.
 - b. A BPML activity is used to make choices about the data.
 - c. A graphical tool creates another layer in your business process.
 - d. A layer allows you to add more details about an activity in the business process.
2. The business process that is checked in to Sterling B2B Integrator is independent of any .bp file that you have saved in a local directory.
 - a. True
 - b. False
3. Which is the language used to create rules in a business process?

(Continued on next page)

Quiz

(Continued)

Questions (Continued)

4. Match the Persistence Level with their appropriate descriptions.

	Persistence Level		Description
1	BP Start Stop (Engine May Override)	a	Choose this level if you do not need tracking data for your process when it is successful and your recovery does not depend on process step data being saved.
2	Step Status Only	b	Choose this level if you know that you do not need persisted status data for successful processing.
3	BP Start Stop Only	c	Choose this level if you need only status information for each step and your recovery needs do not depend on additional data being saved
4	Error Only	d	Choose this level if your recovery needs do not depend on data being saved for the processing steps.

Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Define a business process and the lifecycle of the business process
 - Create a business process
 - Implement a business process
 - Check in a business process model
 - Enable a business process model
 - Manually run a business process model
 - Modify a business process
 - Check out a business process
 - Check in a modified business process
 - Disable a business process model
-

Using the GPM

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Background

Introduction	This lesson provides instructions for using the Graphical Process Modeler (GPM) to create an executable business process.
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Lesson objectives	This lesson is designed to help you to: ■ Define a business process and the life cycle of the business process ■ Create a business process ■ Implement a business process ■ Check in a business process model ■ Enable a business process model ■ Manually run a business process model ■ Check out a business process ■ Modify a business process ■ Check in a modified business process ■ Disable a business process
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Business Processes

What is a Business Process?

Business processes are like road maps. Just as you use a road map to determine the best route through the countryside, Sterling B2B Integrator uses business processes to find the best route through critical business systems, such as purchasing, inventory status, and invoicing.

By analyzing and documenting your business and technical needs, you can determine your business processes. You can then reword those business processes to fit your business needs.

The Lifecycle of a Business Process

This section provides an overview of how Sterling B2B Integrator works. It follows Sterling B2B Integrator through:

- Analyzing a business process
- Defining a business process with services
- Compiling a business process
- Check In a business process
- Starting a business process
- Archiving a business process
- Restoring a business process
- Deleting a business process

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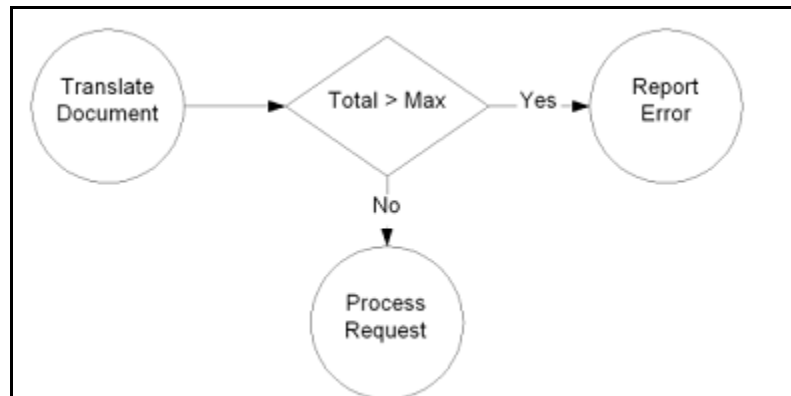
Business Processes

(Continued)

Defining a Business Process

Sterling B2B Integrator is a framework that runs customer-specific business processes. All operations that are carried out on documents and their order is set by an XML-based business process model.

Defining a business process might look like the following diagram. The circles represent activities and the diamond represents a decision point.



Since Sterling B2B Integrator stores business process definitions in XML, there are a couple of ways for a business process analyst to define them:

- Graphical Business Process Modeler included with Sterling B2B Integrator
- Simple text editor

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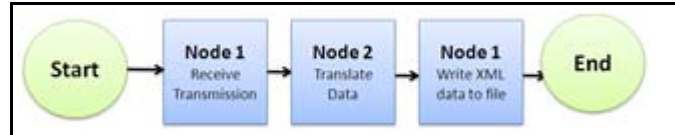
Business Processes

(Continued)

Defining a Business Process (Continued)

Nodes in a business process model

Nodes in a business process model are the graphical representation of the steps the business process goes through as it runs and processes the data involved. The following figure is a simple example.



Services

Flexibility is key to the business analysts who works with the IT staff to determine how business processes needs to be written. Sterling B2B Integrator allows you to write subprocesses with multiple reusable components or as one large service that does a complete process on its own. One way to look at services is to look at where they get the information they process and where they pass it when the processing is complete:

- Internal services are located inside Sterling B2B Integrator; they accept parameters and produce results, but they do not interact with any system outside Sterling B2B Integrator.
- Input services receive data from an outside system.
- Output services send data to an outside system.

(Continued on next page)

Business Processes

(Continued)

Defining a Business Process (Continued)

Adapters

Input and output services are known collectively as adapters. Adapters interact with the “world” outside Sterling B2B Integrator.

Business Process Context

- The service accepts a business process state and produces a modified business process state, or business process context (BPC). BPC also be referred as WFC, or Workflow Context.

Compiling a Business Process

After you create a business process model, you must validate and compile the process before you can run it. Validating ensures that all activities in the business process model are properly configured.

Compiling the business process allows Sterling B2B Integrator to determine the “Start” node ahead of time so business processes are more easily instantiated and run. Since the start pointer for the next activity (node) is stored in the current activity (node) information, compiling the business process also eliminates repeated parsing of XML and reduces the number of queries to the database.

Validating and compiling are done when you validate when saving or using the Validate button inside the GPM. Sterling B2B Integrator will not allow you check-in a non-validated business process.

When writing BPML or copy and pasting bpml text in the Administration interface you will have to validate before being able to move on during the check-in process.

Note: Because a business process is validated does not mean it will run without errors at run-time. It will only verify that the BPML is syntactically correct. It can not verify that the business process will interact with other systems or files correctly.

(Continued on next page)

Business Processes

(Continued)

Checking In a Business Process

Before you can use a business process within Sterling B2B Integrator, you must first check it into the system. You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface. This is the process of adding the Business Process from where you are developing it to the server where it will be executed.

Both the Sterling B2B Integrator interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time that you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

(Continued on next page)

Business Processes

(Continued)

Starting a Business Process

Business processes can be started either manually or automated by the system. Automated can be:

Scheduled- Set to run a specific times or intervals

Invoked- Called or started by another business process.

Bootstrapped- Similar to Invoked, but called from an adapter or event instead of a business process. Such as an adapter monitoring a file folder and kicking off a business process when a file is found.

Dynamic bootstrapping is the selection of the appropriate business process to run when data enters. Input data must enter Sterling B2B Integrator by way of an input adapter. This adapter takes the received data, puts the data and any metadata it finds into an Initial Business Process Context (IBPC), and begins bootstrapping. A business process definition is selected based on the data and/or metadata and instantiated as needed.

Archiving a Business Process

Archiving your business process data not only protects your critical data but also conserves database storage, improving the efficiency of Sterling B2B Integrator.

(Continued on next page)

Business Processes

(Continued)

Restoring a Business Process

After you have archived business process data to an offline location, you can restore that data and view it in Sterling B2B Integrator.

Deleting a Business Process

In Sterling B2B Integrator, you can delete individual business processes (except business processes used by Sterling B2B Integrator). Some reasons that you might delete a business process are:

- Change in Business activities and, as a result, a business process can no longer accomplish the activities.
- You need to remove versions of a business process to reduce the database size.

Note: A business process can not be deleted if there is still a record of an instance in the database. All instances of a bp must be purged from the database otherwise tracking could be impacted.

For example if a bp was executed and is set to not be purged for 3 days, that business process model could not be deleted until 4 days later. This is because there is still an instance record of that bp executing in the database for 3 days after it ran.

(Continued on next page)

Instructor Note: Students do this exercise, walk around, and help where appropriate. Make sure that all students have completed before moving on to connecting the icons in the next exercise.

Business Processes

(Continued)

Creating a Business Process

After analyzing your business needs, determining your business processes, and documenting the results, you must create business process models that represent those business processes. You can then implement and run the business processes with Sterling B2B Integrator.

Important!	Because a business process must be validated to run successfully in Sterling B2B Integrator, you must ensure that you validate the BPML code in each business process before you add the process to Sterling B2B Integrator. The sections in this lesson include steps to validate a business process
-------------------	---

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Instructor Note: Students do this exercise, walk around, and help where appropriate. Make sure that all students have completed before moving on to connecting the icons in the next exercise.

Business Processes

(Continued)

Implementing a Business Process Using the GPM

You can implement a business process in Sterling B2B Integrator with the Graphical Process Modeler (GPM) by following the appropriate steps here:

1. Create a business process model, including:
 - Configuring additional parameters
 - Adding sub flows
 - Assigning rules and conditions
 - Assigning name-value pairs that represent data to locate in process data
 - Configuring the Correlation service to generate correlations for business processes and document exchange
 2. Validate and save the business process model document.
 3. Add the business process to Sterling B2B Integrator.
-

Exercise 5–1 Moving Activities to the Workspace

Scenario You are required to create a business process to check for the quantity of Gizmo items from the inventory XML report. If the quantity is less than the minimum inventory level then the process should display a message to order more items and if the quantity is greater than minimum inventory level then the process should display a message not to order more items.

Instructions **Procedure to Move Activities to Workspace**

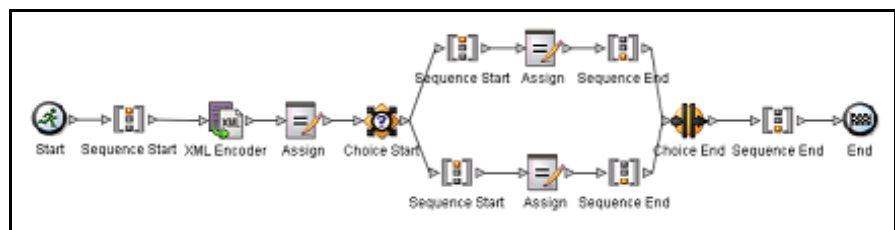
Follow the steps that are listed to move the activities to the workplace.

1. Open the Graphical Process Modeler.
2. Click New.
3. Bring up the Stencils if not already showing.
 - BPML Services
 - All Services
4. Move the following operations and services to your workspace by dragging the icons from the Stencil to the workspace:
 - Start (1)
 - End (1)
 - Sequence Start (3)
 - Sequence End (3)
 - Assign (3)
 - Choice Start (1)
 - Choice End (1)
 - XML Encoder (1)

Exercise 5–2 Connecting Activities

Scenario You need to connect the activities that determine the order in which the activities are run within the Business Process.

- Instructions** Follow the steps that are listed to connect the activities within the business process.
1. Click the triangle at the right of the **Start** node and keep the mouse button down.
 2. Drag the mouse to the left triangle of the **Sequence Start** node and release the mouse button. This creates a link between the nodes.
 3. Connect the activities from the previous exercise as shown in the following figure.



(Continued on next page)

Exercise 5–2 Connecting Activities

(Continued)

Instructions (Continued)

- (Optional) Select **View > Autolayout** to visually align the business process model. Autolayout will evenly space all of the icons. This could move icons off your screen.
- Select **Save** and save the business process model document (.bp file) to your client computer. You do not need to validate this business process model because you have not configured any parameters yet.

Remember!	Give the business process model a name that helps identify the business process.
------------------	--

Important!	If you do validate you will receive an error that ... is not configured or other errors. This is normal as you have not configured anything. You will configure the icons in later lessons.
-------------------	---

- Select a directory on your client computer in the Save dialog box, and name your file **BasicInventoryProcess**, and click **Save**.

You are now ready to configure parameters, add sub flows as necessary, and assign rules and conditions for your business process model. These topics will be covered in the next lesson.

Important!	Certain activities and services require configuration within the GPM. You can perform the configuration with the five element editors in the GPM: Property, Service, Edge, Produce, and Consume.
-------------------	--

Viewing the Element Editor and Configuring Additional Parameters

Example Procedure to Configure Additional Parameters

These are the steps you would take to view an element editor and configure additional parameters.

1. Select an icon on the workspace and click **Edit > Properties** to display the applicable element editor. For example, when you select the Assign icon, the Property editor opens and provides parameters to configure for the Assign activity.
 2. Configure the parameters in the element editor.
 3. Configure the parameters in the applicable element editor for each remaining icon on the workspace as necessary.
 4. Right-click the **Editor** tab and select **Close Editor Tab**.
 5. Validate and save the business process model.
-

Sub-Flow

Subflows in the GPM

When you create a subflow, you create a layer in the business process model when viewed in the GPM. You can expand the view to work with the components of the subflow by using the Layers tab in the Navigation pane.

While a subflow layer provides more design space in the GPM, it does not create a separately-saved business process model. Nor does it create a child business process that is called from a parent.

Subflows are only available in the GPM and are not supported in the actual BPML code. If you were to look at the source code of a business process that contains a subflow you would only see an additional sequence tag. Also if you modify the BPML code manually and then later load in the GPM the subflow information will not show as a separate layer but as a sequence in the main layer.

Exercise 5–3 Adding Sub Flows

Scenario You are required to add a sub flow that creates a layer in a business process model where you can define more details about an activity in a business process. As you design the business process model, identify where sub flows are possible and design them.

Instructions You can follow the steps that are listed to add a sub flow:

1. Select **Add Sub Flow** from the Tools menu. In the Navigation pane, the business process model expands to list the sub flow.
2. Delete the **XML Encoder** and first **Assign** service from the business process by clicking the icon and pressing the delete key.
3. Insert the **sub flow** icon between the **Sequence Start** and the **Choice Start** icons within the business process model.
4. Link the icons.

(Continued on next page)

Exercise 5–3 Adding Sub Flows

(Continued)

Instructions (Continued)

5. Select the **sub flow** icon, right-click, and select **Properties** to display the Property editor to rename the sub flow. Type **SubFlow1** in the Value field. When you click out of the Property editor, the sub flow icon displays the new name.
 6. Click the **sub flow** in the Navigation pane to display a new workspace that includes unlinked Start and End icons.
 7. Create a business process model for the sub flow.
 - a. Add the XML Encoder and Assign icons within the subflow.
 - b. Arrange the icons.
 - c. Link each icon, including the Start and End icons.
 - d. Align the business process model.
 8. Click the parent business process in the Navigation pane (the one that the sub flow is part of) to return to the main process.
 9. Save the business process model.
-

Assigning Rules and Conditions

Introduction

To indicate a decision point in a business process model, you must:

- Assign a rule that describes the decision.
- Provide the conditions under which the decision is true or false.

When running a business process that processes XML documents, you must use XPath to describe rules and conditions for processing the XML document.

Specifically, you must create XPath expressions that:

- Identify particular elements (or nodes) and their positions in both the input and output documents.
- Identify particular elements in the input document for copying into an output document or for further processing.
- Define any constraints of name-value pairs for elements.
- Calculate values that depend on other values.

You can then assign rules that represent the decision to complete activities in the business process. The following table shows sample names and rules for simple syntax:

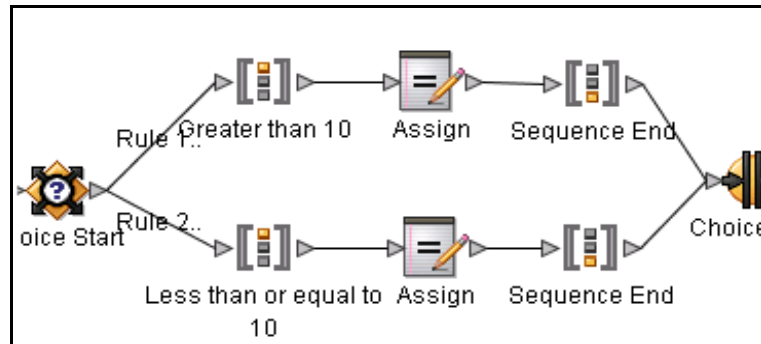
Name	Rule
rule1	X>10
rule2	X<=10
rule3	X<10

(Continued on next page)

Assigning Rules and Conditions

(Continued)**Introduction
(Continued)**

The following figure shows the part of a sample business process model where a decision must occur in a business process. The Choice Start and Choice End icons are included and rule names appear midway through edges (links) in the business process model.

**Rule Manager**

The Rule Manager is a tool in the GPM that is used to create rule conditions that will be evaluated in your business process. A rule is written in XPath and evaluates XML data in Process Data.

Edge Editor

The Edge Editor is used to assign the rules you created to a choice in your bp. Bring up the Edge Editor by double clicking a line coming off of your Choice Start. You must have a rule already created because the Edge Editor only allows you to select a rule and then say if it is true or false. This is why the name of the selected rule appears on the line.

(Continued on next page)

Assigning Rules and Conditions

(Continued)

Creating a Rule

You can create a rule that is defined by simple syntax or complex syntax. For ease of understanding, this procedure is chunked into two parts:

- Create a rule defined by simple syntax
- Create a rule defined by complex syntax using XPath

Create a Rule Defined by Simple Syntax

You can create a rule that is defined by simple syntax in the following manner.

1. Select **Rule Manager** from the Tools menu.
2. Click **Add** and type a name that describes the rule.
3. Type the syntax in the Expression field, and then click **OK**.
4. Repeat steps 1-3 until you finish creating rules that are defined by simple syntax.

(Continued on next page)

Assigning Rules and Conditions

(Continued)

Create a Rule Defined by Complex Syntax using XPath

You can follow the steps listed to create a rule defined by complex syntax using XPath.

1. Select **Rule Manager** from the Tools menu.
 2. Click **Add** and type a name that describes the rule.
 3. Click the button to the right of the Expression field to display the XPath Expression Builder.
 4. Display the expression type that you want to perform, double-click the function to add it to the Expression field, and then type values to complete the expression type as needed.
 5. Display the token set attribute to complete the function of the expression type and double-click the attribute to add it to the Expression field.
 6. Repeat steps 1-5 until you finish defining this XPath expression and click **OK**.
 7. Repeat steps 3-6 until you finish creating rules that are defined by complex XPath.
-

Exercise 5–4 Using Rules

Scenario	The rule manager allows for the definition of expressions that evaluate to true and false. The rules can be used for conditional execution of activities within the Business Process.
-----------------	---

Instructions	You can follow the steps that are listed to use a rule. 1. Click Tools > Rule Manager from the GPM main menu. 2. Click Add to bring up the Rule Editor. 3. Populate the Name box with Inventory Count. 4. Type the following string in the Expression box: <code>count(INVENTORY/PRODUCT) > 10</code> 5. Click OK in to add the rule to the list of rules in the Rule Manager. 6. Click OK on the Rule Manager window to save the rules.
---------------------	--

Exercise 5–5 Assigning a Rule

Scenario	You are required to assign rules and define conditions.
-----------------	---

-
- | | |
|---------------------|--|
| Instructions | <p>You can follow the listed steps to assign rules and define conditions.</p> <ol style="list-style-type: none">1. Click the edge (link) between the Choice Start and the Sequence Start icon on the top to display the Edge editor.2. Click Add to display the name and value columns in the Edge editor.3. Click in the field and select Inventory Count under the Name column.4. Select True for the selected rule under the Value column. When you click out of the Edge editor, you see the rule name on the edge.5. Click the edge (link) between Choice Start and the icon instance on the bottom to display the Edge editor.6. Click Add to display the name and value columns in the Edge editor.7. Click in the field and select Inventory Count under the Name column.8. Select Not true for the selected rule under the Value column. When you click out of the Edge editor, you see the rule name on the edge.9. Save the business process model. Do not Validate during the Save. |
|---------------------|--|
-

(Continued on next page)

Exercise 5–6 Viewing and Configuring the Different Element Properties

Scenario You are required to view and configure the different element properties.

- Instructions**
1. Double-click the first **Sequence Start** operation in the model to open the Property Editor and type the following text in the Value column for the Name field: **Main Sequence**.
 2. Click **Sub Flow** in the Navigation Pane.
 3. Click the **Assign** activity that immediately follows XMLEncoder in the subprocess and enter the following information:

Name	Value
Constant	N/A
To	TotalPrice

4. Click the **XMLEncoder** Service to open the Service Editor and complete the following:
 - Select **XMLEncoder** from the Config drop-down list box.
 - Select **Use existing XML document** from the drop-down list box in the Value column for the **mode** field.
 - Select **Yes** from the drop-down list box in the Value column for the **output_to_process_data** field.
5. Click **BasicInventoryProcess** from the Navigation Pane, so the BasicInventoryProcess BP is displayed on the workspace.
6. Click the **TOP** Sequence Start and type the following text in the Value column for the Name field: Greater than 10.

(Continued on next page)

Exercise 5–6 Viewing and Configuring the Different Element Properties (Continued)

**Instructions
(Continued)**

7. Click the **BOTTOM** Sequence Start and type the following text in the Value column for the Name field: Not Greater than 10.
8. Click the **Assign** activity (Greater than 10) and enter the following information:

Name	Value
Constant	Do Not Order More Items
To	Result

9. Click the **Assign** activity (Not Greater than 10) and enter the following information:

Name	Value
Constant	Order More Items
To	Result

Exercise 5–7 Saving and Validating a Business Process

Scenario Validating a Business Process ensures that all activities are completed properly and the BPML generated from the graphic is syntactically correct. The output window at the bottom of the Graphical Process Modeler shows any error messages. Saving without validating saves the current graph so that you can open it in the GPM later. Without validation, you cannot execute the Business Process.

Instructions You can follow the steps that are listed to save and validate a business process.

1. Select **File > Save**.
2. Click **Yes** when asked to validate the process model upon saving.
3. Complete the **Save** dialog:
 - a. Select the folder location.
 - b. In the FileName field, enter BasicInventoryProcess.
 - c. Select **OK**.
4. If, the process model is not valid the output window appears at the bottom of the Graphical Process Modeler with a display of the error messages. Correct the errors and try saving the Business Process again.

Implementing a Business Process

Introduction

After creating or modifying a business process model, you must validate, check in, and test it before running it in Sterling B2B Integrator. Use a test environment for checking in and testing your processes before you run them in your production environment.

The first time that you check in a business process, you create a business process definition; a uniquely named copy of the process model stored in Sterling B2B Integrator. The business process model that is stored in Sterling B2B Integrator is independent of any .bp file of the process that you have saved in a local directory.

At any time, you can modify an existing process model and check in a new version to Sterling B2B Integrator, or save the modified copy of the process in Sterling B2B Integrator as a new process altogether. Version management is an important part of business process maintenance.

Checking in a Business Process

You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface.

Both the Sterling B2B Integrator interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time that you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

The check in procedure differs depending on your goal and the application you are using.

(Continued on next page)

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in

During the check-in procedure, Sterling B2B Integrator prompts you to select various processing options for a business process model. These options are related to archiving, document storage and tracking, persistence levels and other instructions that are related to your business needs. The settings that you choose can affect system performance. Before you check in a business process model, have an idea of what the appropriate selections are for your installation.

The following table describes the options:

Field	Description
Process Type	Type a name that identifies the business process model. This is the business process definition name that is stored in Sterling B2B Integrator.
Description	Type a description that helps identify the business process that you are creating. This is stored in Sterling B2B Integrator with the business process definition you are checking in. If the process model is an updated version of a business process, you may want to use a meaningful version number or some other content to identify the version.
Continued on next page	

Implementing a Business Process (Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Set Queue	Sterling B2B Integrator allows you to set performance optimizations by queue, defining queue levels to allocate resources. Select the previously allocated queue level that you want for this business process model for processing.
Use BP Queuing	If selected, the bp is placed in a bp queue for processing. This is the default and recommended method. To run the bp in BP in Non-BP Queued mode, clear the check box. In BP Non-Queued mode, however, the Persistence Level for the BP is zero by default, so the process cannot be manually restarted or resumed. Also some services are unable to operate with zero persistence
Enable Transaction	Select this option to instruct Application to treat the entire process as a single transaction so that either all of the steps complete, or, in the event of an error, none of them do. When an error occurs, no data is committed; data returns to its pre-process state. Note: Enabling transactions applies only to services that support transaction mode. See the reference information for specific services to determine whether this option is supported. By default, this transaction mode is not enabled. Note: If transaction management is already built into the process model (the model includes Start and End Transaction services), do not select this option or the process will fail.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Commit All Steps When There Is An Error	Select this option to have the business process commit all work to the database, at the time an error is generated. With this option, you can either roll back or commit all work before the error in the onfault block is called. If you select this option, all steps before the error are committed to database and the business process keeps running on the onfault block path. If you do not select this option, all steps before the error will be rolled back and the onfault block path is called. By default, this option is not enabled. Note: This option is valid when the BPML is configured with an onfault block and Enable Transaction is selected
Category	You can optionally enter a category name to which this process model belongs, creating the category. The category does not affect processing; categories are for future product enhancements.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Persistence Level	Select the level of data to retain for generating a status report that describes each step that the business process completes. The levels are: ■ Full - Retain all data for this business process, including associated documents, activities, and all process data. ■ Step Status (Engine May Override) - The recommended persistence level for most process models because it helps optimize performance while providing a level of recovery suitable for most business needs. Persists process data and documents for service steps according to the persistence level supported by the service in the step. ■ BP Start Stop (Engine May Override) - Choose this level if you know that you do not need persisted status data for successful processing. Because service level settings override the selection, you may have additional data in a recovery scenario. * Persists process data and documents for service steps according to the persistence level supported by the service in the step. ■ System Default - Assume that for the data, configuration is already defined in Sterling B2B Integrator to retain data. Persists data according to the global persistence setting configured in the properties file. ■ Step Status Only - Choose this level if you need only status information for each step and your recovery needs do not depend on additional data being saved. ■ BP Start Stop Only - Choose this level if your recovery needs do not depend on data being saved for the processing steps.
Continued on next page	

Implementing a Business Process

(Continued)

**Determining
Processing Options
During Check-in
(Continued)**

Field	Description
	■ Zero - This level does not persist any business process data for recovery or process tracking. Choose this level only if you are certain that you will not need the data. Note: Zero persistence is available only if the process is running in Synchronous start mode. ■ Error Only - Choose this level if you do not need tracking data for your process when it is successful and your recovery does not depend on process step data being saved. ■ BP Start Stop Only (No Errors) - The first step of a business process is persisted in full, and last step is persisted in minimal. However, when the business process runs into an error which makes it stop running, no error step is persisted. The business process stays in ACTIVE state. If the process runs into an error, and there is an onfault step in the business process, the business process will complete. ■ Override None No IC - Does not retain any details while the business process is running or in the current process, and will not retain any details after the business process is run.
Event Reporting Level	Select the level of event reporting that you would like to retrieve for this business process when it runs. The levels are: ■ Full - Generate events for the business process, including the business process start and end time, start and end times for all services or adapters running as a result of this business processes, and any resulting errors and exceptions. ■ Minimal - Generate events for the business process, including the business process start and end time and any resulting errors and exceptions. ■ None - Do not generate any event reporting.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Recovery Level	Select the level of recovery for this business process if the business process should halt during execution. The levels are: ■ Auto Resume - Resume the business process at the point at which the business process halts. ■ Auto Restart - Restart the business process from the beginning. ■ Terminate - Terminate the business process. ■ Manual - Require you to resume or restart the business process manually.
Document Storage	Select the level of document storage for documents that process when the business process runs. The levels are: ■ Database - Store documents in the database. ■ File system - Store documents in the file system. ■ Archiving and purging from the file system may require special handling. ■ System Default - Store documents in the file system or database, according to how you configured archiving and purging in Sterling B2B Integrator. ■ Inherited - If this is a subprocess, use the same storage type indicated for the parent business process.
Continued on next page	

Implementing a Business Process (Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Enable BP Execution Setting	This option displays only when you are using a clustered Application environment. Select this option to display the available choices for specifying the node to run the process on. If you do not specify a node, the system uses any available node. To select a node: 1. In the Select Node field, select the node on which you want the process to run. 2. In the Execution Node Specific field, specify whether the process must run on the selected node or can use others. Options are: ■ Preferred Node - Run the process on the selected node unless the node is unavailable (inactive for some reason). ■ Mandatory Node - Always run the process on the selected node (if the node is unavailable for some reason, the process will fail). Note: The Execution Role Specific field is reserved for future use.
Deadline Setting	Deadline setting indicates how much time the process may take to complete. Select the appropriate setting. Settings are: ■ Do not set deadline - Allows you to run the business process without a deadline. ■ Complete by Deadline - Allows you to specify a deadline time, in hours and minutes, by which the business process must complete process once it starts. ■ First Notification: Hours and Minutes - Allows you to specify whether to receive notification before a business process deadline. ■ Second Notification: Hours and Minutes - Allows you to specify whether to receive another notification before a business process deadline.
Continued on next page	

Implementing a Business Process

(Continued)

Determining Processing Options During Check-in (Continued)

Field	Description
Document Tracking	Enable or disable this selection. Select this option if you want the documents that are involved in this process model to be trackable according to the configured document tracking settings. Document tracking at the business process level is disabled by default, and this setting overrides the global document tracking setting. If you are not using Application document tracking features, leave the option disabled, to reserve system resources for other tasks.
Set OnFault Processing	Enable or disable this selection. Select this option to enable the on-fault activity that is specified in the process to immediately execute in the event of a system error.
Life Span	Use these options to: Indicate the length of time that data is pertaining to each instance of this business process model will remain in the system after the process is completed, to be available for monitoring, tracking, and reporting activities. Specify whether to archive or purge the process-related data when the life span expires. ■ Process Specific - Number of days or hours, or days and hours combined, that data for each instance of this business process model are permitted to remain in the system. ■ Expired Business Processes should be - Select either Archived or Purged for expired data. Archived data is stored in your file system. ■ System Default - Select this option if you want the data for instances of this business process model to expire and be archived after two days. This option is configurable.

Implementing a Business Process

(Continued)

Enabling Document Tracking

Enabling Document Tracking makes the document activity for this business process available from the Advanced Search menu. Access Document Tracking from **Business Process > Monitor > Advanced Search > Documents**.

Setting onFault Processing

The onFault element is used to handle errors. You can include onFault elements in any complex activity for which it may be necessary to recover from faults so that the process can continue. The onFault element contains activities for handling faults.

Setting the Persistence Level

The persistence level options allow you to specify, at the business process level, the types of data to retain in the database for the process whenever it runs. Persisted data can be crucial to process recovery in the event of errors and is useful for monitoring and tracking activities, including generating status reports. Persisted data includes any of the following types of data. Status information for the steps in the process is composed of:

- General status information for the steps (start and stop time, step name)
- Error information for failed steps
- Data describing the start and stop conditions of the process
- Process data
- Documents that are associated with the process

The types of data persisted for a process depends on the activities the process involves (such as whether the process has documents that are associated with it) and your persistence level settings.

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Implementing a Business Process

(Continued)

Setting the Persistence Level (Continued)

Persistence can also be set globally (at the system level) for all processes; by default it is set to FULL persistence, although it may have been customized for your installation. If your installation uses the default FULL setting for the system, you generally do not need to change persistence at the business process level unless you are seeking performance improvements. For the process model you are checking in, your selection here overrides the global persistence level that is set in the properties file.

Important!	Two exceptions might affect your persistence selection. First, remember that some services support only a limited selection of persistence settings, regardless of the level of persistence you assign to the process model. Second, any persistence overrides configured in the BPML for steps in the process model override the setting that you make here for the specified step or steps.
-------------------	---

Choose the option that provides the type of data you need for monitoring, tracking, and recovery in the event of a failure. Remember that persisting data can slow system processing by filling up the database. Consider your archive and purge settings in addition to the amount and type of data you intend to persist.

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Implementing a Business Process

(Continued)

Choosing Document Storage

Document storage type is the storage type that you want for documents that process when the business process runs. The levels are:

- **Database** - Store documents in the database.
- **File system** - Store documents in the file system. This option can be best for large documents. Using this method, you must handle archiving and purging the documents from your file system.
- **System Default** - Store documents in the file system or database, according to how you configured archiving and purging in Sterling B2B Integrator.
- **Inherited** - If it is a subprocess, use the same storage type indicated for the parent business process.

When you check in a business process, Sterling B2B Integrator gives you the opportunity to select the location for storing the business process.

Setting the Process Instance Life-Span

When you check in a business process, Sterling B2B Integrator gives you the opportunity to set the Life Span of specific instances generated by that business process. The Life Span setting tells Sterling B2B Integrator how long the instance should remain in the active system. In addition, when the business process instances expire, you can set them to either automatically archive or purge themselves from the system.

(Continued on next page)

Implementing a Business Process

(Continued)

Setting the Expired Characteristics of a Process Instance

When you check in a business process, Sterling B2B Integrator gives you the opportunity to determine whether the specific instance of the business process needs to be Archived or Purged. This must be set up in the business process. Archive moves instances of business processes that have been run and any associated data from the live system to a storage area. Once in the storage area the business process instance (and data) is available to be moved to offline media for storage. Purge permanently removes business process instances that have run and any associated data. Archive and purge help to limit the amount of information that is kept in the live system. It is important to note, data that must be kept long term should be archived and stored in a safe media.

Enabling a Business Process

You must enable a business process that is checked in to the system before it can be available to run. Enabling a process model tells the system that this process is available to run. You cannot enable a business process unless it was validated at check-in. Business processes checked in through the Administration UI are automatically validated.

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Implementing a Business Process

(Continued)

Creating Permissions

Select this option to have a permission selection created for this business process model automatically. The permission can be assigned to user accounts to allow or restrict user access to running the process. Permissions are optional. If none are assigned, the process is available for all users.

Setting the Default Version of a Business Process

When you check a process model into the system, you can also set it as the default process. The default process is the process version that Sterling B2B Integrator uses when running a request of a new process instance that does not have explicit version information. The detail information about the versions of business processes is discussed in the later part of this lesson.

Disabling a Business Process

Only business processes set as the default version can run. A business process that was checked in to the system can be disabled, or made not available for execution, if it is not set as the default version. When a business process version is disabled, the system will not start new instances of the process.

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Implementing a Business Process

(Continued)

Check in the Business Process

You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface.

Both the Administration Interface or User Interface and the GPM provide a Source Manager feature that you use to check in a business process. The two Source Managers differ slightly, but both allow you to check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time when you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

There are three ways to check in a business process. The Business Process Manager provides two different methods. You can check in a business process by using either the:

- Graphical Process Modeler
 - Business Process Manager
-

Exercise 5–8 Check in a Business Process using Graphical Process Modeler

Scenario You are required to check in a business process using GPM.

Instructions You can follow the steps that are listed to check in a business process with GPM.

Important!	If you are not already in the Graphical Process Modeler, complete steps 1 and 2. Otherwise, skip to step 3.
-------------------	---

1. Click **Business Process > Manager** on the Sterling B2B Integrator Administration menu bar.
2. Click Go! on the Sterling B2B Integrator Business Process Manager page, and click next to **Graphical Modeling: Run Graphical Process Modeler**.
3. Click **Tools > Source Manager** in the Graphical Process Modeler.
4. Log on to the Source Manager with the Sterling B2B Integrator login and password.
5. Click **Process > Add**.
6. Type the Process Name **BasicInventoryProcess** and click **Next**.

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Exercise 5–8 Check in a Business Process using Graphical Process Modeler (Continued)

Instructions (Continued)

7. Browse out and select your BP file using the **Browse** Button.
8. Type **Basic Inventory BP** as the description. Click **Next**.

Important!	For the following fields, your view may differ, depending on whether you specified Expert Mode for the Source Manager through the GPM. If you did not specify Expert Mode, click Next to display the following fields.
-------------------	--

9. Set the Queue to **4**, the Persistence Level to **System Default**, the Event Reporting Level to **None** and the Recovery Level to **Manual**. Set the Document Storage to **System Default** and click **Next**.
10. Check the option, **Do Not Set Deadline**.
11. Accept the System Defaults for Archive on the next two screens by clicking **Next**.
12. Check **Validate**.
13. Check **Enable Process**.
14. Do not check **Enable Document Tracking** and **Set onFault Processing**.
15. Click **Finish**.
16. Click **OK**.

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Exercise 5–9 Checking in a Business Process using Business Process Manager

Scenario You are required to check in a business process using the business process manager.

- Instructions** You can follow the steps that are listed to check in a business process using the business process manager:
1. Go to **Business Process > Manager** from the Sterling B2B Integrator Administration menu.
 2. Go to **Create Process Definition** and click **Go!**
 3. Type the Process Name **BasicInventoryProcess1**.
 4. Click the radio button for **Check in Business Process created by the graphical modeling tool** under Select an input mode for defining the new process and click **Next**.
 5. Click **Browse** to locate the bp file on the Process Check-in screen and then click **Open**.
 6. Type a **description**, and then click **Next**.
 7. Uncheck **Document Tracking** and **onfault processing** on the Process Levels screen, set the Priority Level to **4**, the Persistence Level to **System Default**, the Recovery Level to **Manual**, the Event Reporting Level to **None** and the Document Storage to **System Default**. Click **Next**.
 8. Leave the amount of time set to **System Default** on the Life-Span Screen.
 9. Click **Next**.
 10. Click **Finish** on the Process Confirm screen.

Important!	The Enable Business Process checkbox is checked.
-------------------	--

Exercise 5–10 Run a Business Process Manually

Scenario Sterling B2B Integrator runs most business processes automatically. There are times however; when it is necessary to run a business process manually. This exercise describes the procedure for manually running a business process. When you build a new business process, it is important to test it before moving it into production. You can do this by manually running the business process.

Instructions You can follow the steps that are listed to manually run a business process.

1. Click **Business Process > Manager**.

The Business Process Manager screen will be appears.



(Continued on next page)

Exercise 5–10 Run a Business Process Manually (Continued)

**Instructions
(Continued)**

2. Type **BasicInventoryProcess** in the Search text box and click **Go!**
 3. Click the **Execution Manager** icon for BasicInventoryProcess.
 4. Click the **execute** icon.
 5. Click **Browse** and go to **Class Labfiles** to locate the test data. Your instructor will provide the details of the location of the lab files.
 6. Locate the file **InventoryOf3.xml** in the Class labfiles located in the Datafiles folder on the Desktop, and click **Open**.
 7. Click **Go!** Observe the separate Execute Business Process window while the business process runs.
 8. Close the **Execute Business Process** window when the Business Process has completed running.
 9. Repeat steps 4-9 using **InventoryOf11.xml** as the input document.
 10. Click **Return**.
-

Exercise 5–11 Obtaining Results

Scenario Follow along to see the record of your executed business process instance data. We'll learn more about tracking business process instance data later in this course.

Instructions Follow the steps that are listed to view the record of your executed business process instance data.

1. Click **Business Process > Monitor > Current Processes**.

The Business Process Monitor screen displays the current processes.



The screenshot shows the IBM Sterling B2B Integrator interface. The top navigation bar includes links for Admin Console, Channels, Operator, Community Management, and Advanced File Transfer. The left sidebar contains an Administration Menu with options like Business Processes, Monitor, and Trading Partner. The main content area is titled 'Monitor' and displays a table of business process instances. The table has columns for Status, ID, Name, State, Started, Ended, Deadline, Parent/Child, and Ex. Two instances are listed, both with a status of 'Completed' and a name of 'BasicInventoryProcess'. The first instance has a started time of 12:26:52 PM and an ended time of 12:26:53 PM. The second instance has a started time of 12:25:19 PM and an ended time of 12:25:20 PM. A checkbox for 'Automatically refresh every minute' is checked. Below the table, it says 'Last update on 03/02/2011 10:19:26 AM'.

Status	ID	Name	State	Started	Ended	Deadline	Parent/Child	Ex
Completed	12656	BasicInventoryProcess	Completed	3/04/05 12:26:52 PM	3/04/05 12:26:53 PM	None	None	1
Completed	12650	BasicInventoryProcess	Completed	3/04/05 12:25:19 PM	3/04/05 12:25:20 PM	None	None	1

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Exercise 5–11 Obtaining Results

(Continued)

Instructions (Continued)

2. Locate first instance of **BasicInventoryProcess** in the list.
 3. Click corresponding **Business Process ID** to view the Business Process Detail page.
 4. Click the **Info** icon in the Document column of step 2, the Assign Service.
 5. Click the **Info** icon in the Instance Data column of step 3.
 6. Click the **Info** icon in the Instance Data column of step 4.
 7. Click **Return**.
 8. Repeat steps 2-6 with second instance of **BasicInventoryProcess**.
 9. Click **Return** when complete.
-

Modifying a Business Process

Introduction

To modify or reuse a business process that is currently checked in to Sterling B2B Integrator:

1. Check out the business process.

The check-out process allows you to save the business process to your client computer. You can then open the checked-out business process file in the GPM and modify the business process model.

2. Modify or Edit the business process as needed.
3. Check the new version in to Sterling B2B Integrator when finished.

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Modifying a Business Process

(Continued)

Checking Out a Business Process

When you need to modify or reuse a business process that is currently checked in to Sterling B2B Integrator and enabled for modification, you can check out the business process from Sterling B2B Integrator. The check-out process allows you to save the business process to your client computer. You can then modify the business process document to create new versions of a business process or a new business process. If you have logged in to Sterling B2B Integrator using Source Manager in the GPM, you can check out a business process using Source Manager. Source Manager also allows you to check out one or many business processes at the same time.

You may want to check out a business process when:

- A business analyst creates or modifies the business processes and has limited access to the host.
- An employee modifies the business processes from a remote location.
- A business process exists on a test environment and is ready for use in the production environment.
- Sterling B2B Integrator checks out a copy of the selected process version, not the original version.

If the selected business process is enabled, it remains enabled until it is replaced by a newer version.

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Modifying a Business Process

(Continued)

Making Changes to an Existing Business Process Model

To change a process model that was checked into Sterling B2B Integrator, you first need to check out the process. This locks the process so that no other users will make simultaneous changes. The checked out version of a process model remains locked to other users until that user checks in the process.

The GPM is the preferred method to change a business process model, although advanced users prefer to change using BPML. When you choose to work in the BPML, and then reopen the process model in the GPM, it will open the process using the default layout options. Since the default layout can appear differently than your specified layout, it is recommend to work in the process modeler.

Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler

Instructions

You can use the following procedure to edit the particular business process.

1. Click **Tools > Source Manager** in the Sterling B2B Integrator Graphical Process Modeler.
2. Log on to the Source Manager with your Sterling B2B Integrator login and password.
3. Scroll through the list of business processes until you find **BasicInventoryProcess**.
4. Select **BasicInventoryProcess** process and click the **Check out a Process** icon.
5. Click **OK** to accept the location where BasicInventoryProcess.bp will be saved.

Important!	The Lock check box indicates that a user is operating on the business process.
-------------------	--

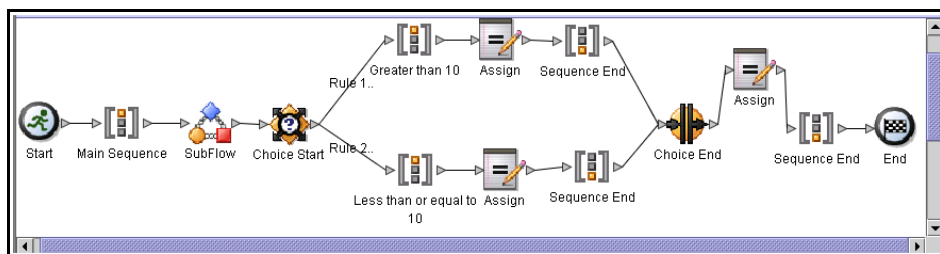
6. Click **Open file** icon in the Sterling B2B Integrator Graphical Process Modeler, and select the location where **BasicInventoryProcess** was saved.

(Continued on next page)

Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler (Continued)

Instructions (Continued)

7. Modify the business process model by adding **Assign** BPML activity.
 - Click and drag Assign to the workspace from the BPML Operations stencil.
 - Break the link after Choice End and before Sequence end.
 - Connect the new Assign to Choice End and the Sequence End respectively.



8. Right-click the **Assign** activity, then click **Properties**.
9. In the Property Editor:
 - Type **count(INVENTORY/PRODUCT)** in the **from** field
 - Type **TotalCount** in the **to** field

Important!

The **to** field cannot have any spaces.

10. Click **File > Save**.

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Exercise 5–12 Editing a Business Process Using the Graphical Process Modeler (Continued)

**Instructions
(Continued)**

11. Click **Yes** on the Validate on save window.
12. Check in the process model.
13. Run the process model with either Inventory documents.

Important!	Be certain to validate again and enable during check-in; otherwise you cannot run the process.
-------------------	--

14. Observe **Info** icon of the Instance Data column of step 4.
-

Exercise 5–13 Editing a Business Process Using BPML

Instructions

You can use the following procedure to edit a particular business process.

1. Click **Business Process > Manager** on the Sterling B2B Integrator Administration page.
2. Type **BasicInventoryProcess** in the Search Process Name field.
3. Click **Go!**
4. Click the **Source Manager** icon for the BasicInventoryProcess business process definition.
5. Click the **Edit** icon for the version of business model you are planning to edit.
6. Type a **Description** for versioning.
7. Change the following line:

```
<condition>count (INVENTORY/PRODUCT) &gt; 10</condition>
```

to the following:

```
<condition>count (INVENTORY/PRODUCT) &gt; 15</condition>
```

(Continued on next page)

Exercise 5–13 Editing a Business Process Using BPML

(Continued)

8. Click **Validate** and click **Next**.
 9. Accept the Process Levels defaults. Click **Next**.
 10. Accept the default settings on the Deadline settings screen. Click **Next**.
 11. Accept the default settings on the Life Span screen. Click **Next**.
 12. Select the current version as the **Default**. Click **Next**.
 13. Click **Finish**.
 14. Run the process model with **InventoryOf11.xml** document.
 15. Observe the process. It now instructs you to “Order More Items”.
-

Exercise 5–14 Checking out the Business Process with Edited BPML

- | | |
|---------------------|---|
| Instructions | <hr/> <ol style="list-style-type: none">1. Select the BasicInventoryProcess in Source Manager, and select Process > Check out.2. Go to your working folder on your local system when prompted to save the process to your client computer, and click Save.3. Return to the GPM when the save is complete, and select File > Open.<ul style="list-style-type: none">■ Go to your working folder , and choose BasicInventoryProcess. Click Open. Notice the changes to the names of the Sequence Starts.4. Click Tools > Rule Manager.5. Click your rule, and select Edit. <p>Note that your rule now test for 15 instead of 10.</p> <hr/> |
|---------------------|---|

Version Control

Versioning

Sterling B2B Integrator maintains a record of each business process, when it is checked in, when it is checked out, and when it is modified. This simple process keeps two users from inadvertently overwriting each other's modified processes.

As each business process definition is checked into the system, a versioning record begins. Furthermore, as a different version of a process is checked in, you specify the default version of that process. This is how Sterling B2B Integrator knows which version of a process to run when new data enters the system or when a specific version is not explicitly requested.

Important!	You must also validate and enable the business process upon check-in for that version to become the default.
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The Execution Manager allows you to manually select which version of a business process to execute. This is useful when you are building a new version of business process and want to compare results. In addition, Sterling B2B Integrator can then keep track of which version of a process was running for a particular business process instance.

Important!	It is recommended that you use the description for setting the versions.
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Exercise 5–15 Viewing a Business Process Definition Version from within the GPM Source Manager

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| Instructions | <hr/> <p>Use the following procedure to examine the version information for a particular business process.</p> <ol style="list-style-type: none">1. Click Tools > Source Manager in the Sterling B2B Integrator Graphical Process Modeler.2. Log on to the Source Manager.3. Scroll through the list of business processes until you find BasicInventoryProcess.4. Expand the view of BasicInventoryProcess to see the list of versions for this business process definition.5. Close the Source Manager. <hr/> |
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Exercise 5–16 Viewing a Business Process Definition Version from within Sterling B2B Integrator

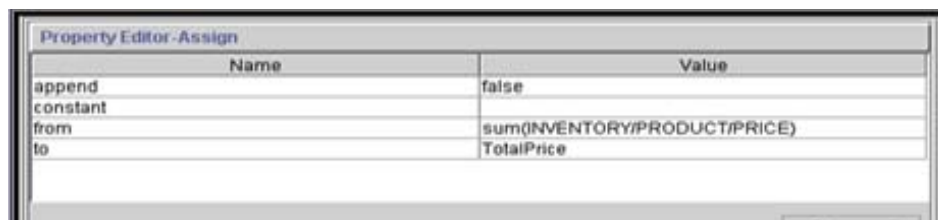
Instructions	Use the following procedure to examine the version information for a particular business process. 1. Click Business Process > Manager on the Sterling B2B Integrator Administration home page. 2. Type BasicInventoryProcess in the Search Process Name field. 3. Click Go! 4. Click the Source Manager icon for the BasicInventoryProcess business process definition. 5. Observe the version information in the Source Manager; specifically the Date and Change History columns.
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Exercise 5–17 Completing the Process

Instructions

In this exercise, an XPath expression needs to be added in the sub flow within the BasicInventoryProcess bp .

1. Open the GPM, if not already open.
2. Click **Tools > Source Manager** in the Sterling B2B Integrator Graphical Process Modeler.
3. Log on to the Source Manager with your Sterling B2B Integrator login and password.
4. Scroll through the list of business processes until you find **BasicInventoryProcess**.
5. Select **BasicInventoryProcess** process and click the **Check out a Process** icon.
6. Click **OK** to accept the location for where BasicInventoryProcess.bp was saved.
7. Click **Open file** icon in the GPM and select the location of where BasicInventoryProcess.bp was saved.
8. Click the **Sub Flow** in the Navigation Pane. Click the **Assign** activity in the model to open the Property Editor for the operation.
9. Remove the text in the Value column for the Constant field.
10. Type the following XPath expression text in the Value column for the From field: sum (INVENTORY/PRODUCT/PRICE).



Name	Value
append	false
constant	
from	sum(INVENTORY/PRODUCT/PRICE)
to	TotalPrice

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Exercise 5–17 Completing the Process (Continued)

**Instructions
(Continued)**

11. Click **File > Save**.
12. Click **Yes** on the Validate on save window. You should receive a success message where your editors have displayed.
13. Check in the process model.
14. Run the process model with either Inventory documents.
15. Obtain the results from running the business process.

If you encounter any errors or have questions on this exercise, contact your instructor.

Important!	Be certain to validate again and enable during check-in, otherwise you cannot run the process.
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Quiz

Questions

1. Which of the following statements are true regarding subflows? Select all correct answers.
 - a. An internal service is used to call other business processes.
 - b. A BPML activity is used to make choices about the data.
 - c. A graphical tool creates another layer in your business process.
 - d. A layer allows you to add more details about an activity in the business process.
2. The business process that is checked in to Sterling B2B Integrator is independent of any .bp file that you have saved in a local directory.
 - a. True
 - b. False
3. Which is the language used to create rules in a business process?

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Quiz

(Continued)

Questions (Continued)

4. Match the Persistence Level with their appropriate descriptions.

	Persistence Level		Description
1	BP Start Stop (Engine May Override)	a	Choose this level if you do not need tracking data for your process when it is successful and your recovery does not depend on process step data being saved.
2	Step Status Only	b	Choose this level if you know that you do not need persisted status data for successful processing.
3	BP Start Stop Only	c	Choose this level if you need only status information for each step and your recovery needs do not depend on additional data being saved
4	Error Only	d	Choose this level if your recovery needs do not depend on data being saved for the processing steps.

Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Define a business process and the lifecycle of the business process
 - Create a business process
 - Implement a business process
 - Check in a business process model
 - Enable a business process model
 - Manually run a business process model
 - Modify a business process
 - Check out a business process
 - Check in a modified business process
 - Disable a business process model
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